

CHY1008

Chemistry and Environmental Studies

Module 2

Part 4

Chemical Kinetics & Catalysis

States of Matter, Thermodynamics, Kinetics and Catalysis

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Chemical Kinetics

Introduction

- In *Thermodynamics*, we learnt **why/when** a chemical reaction occurs
- In *Chemical kinetics*, we will learn
 - **How fast** a reaction proceeds at a given moment?
 - What will reactant and product **concentrations** be when the reaction is complete?
 - **Will the reaction proceed** by itself and release energy, or will it require energy to proceed?



Reactions occur
at a wide range
of rates!

An **explosion (A)** is much faster than the process of ripening **(B)**, which is much faster than the process of rusting **(C)**, which is much faster than the process of human aging **(D)**.



Chemical Kinetics

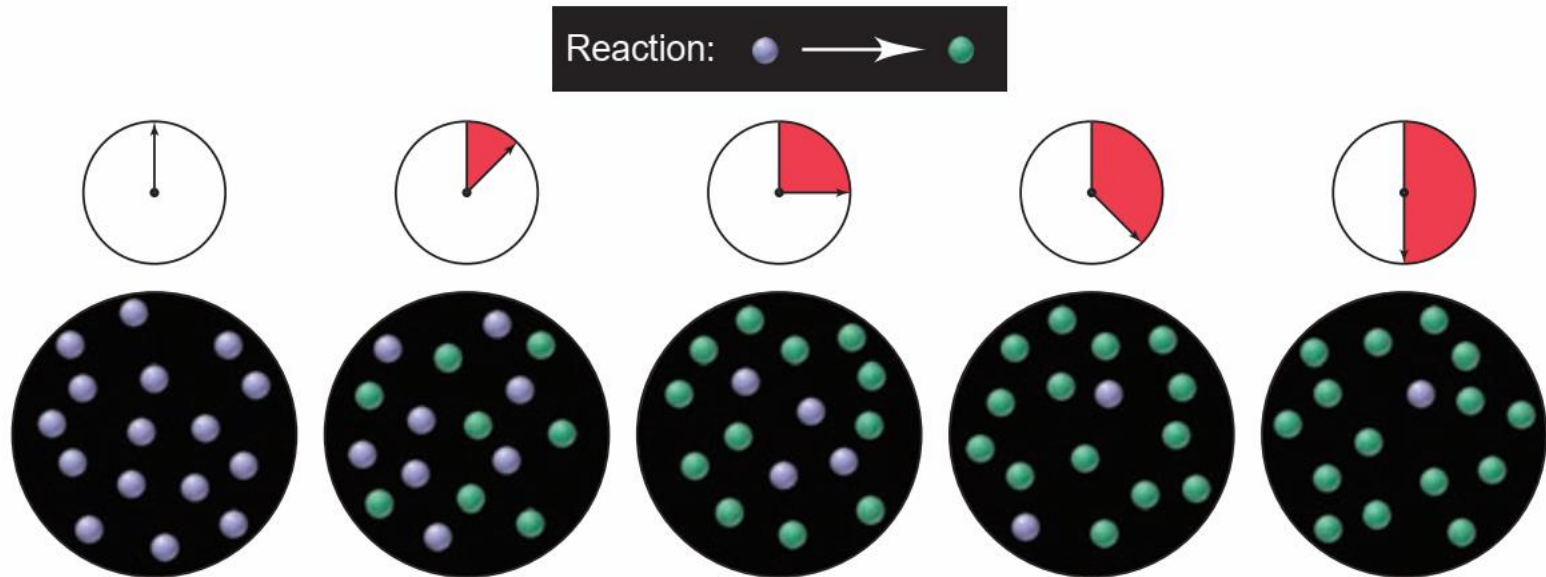
Chemical kinetics, also known as reaction **kinetics**, the branch of physical **chemistry** that is concerned with *understanding the rates of chemical reactions*.

Equilibrium and Chemical Kinetics

Chemical equilibrium is concerned with composition of reaction mixture *at equilibrium*.

Chemical kinetics deals with the composition of reaction mixture/reaction that is *far away from equilibrium*.

Hence, kinetics is an important tool in real life situations.



Chemical kinetics is the study of *reaction rates*, the changes in **concentrations** of reactants (or products) as a function of **time**.

Rate of a reaction

How do we define the rate of the reaction quantitatively?



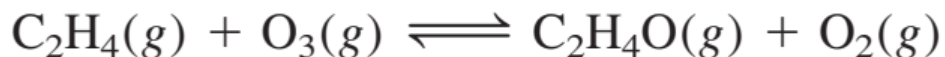
$$\text{Rate of reaction} = -\frac{\text{change in concentration of A}}{\text{change in time}} = -\frac{\text{conc } A_2 - \text{conc } A_1}{t_2 - t_1} = -\frac{\Delta(\text{conc } A)}{\Delta t}$$

$$\text{Rate} = -\frac{\Delta[A]}{\Delta t}$$

Rate expression for the decomposition of ozone,



Rate of a reaction



$$\text{Rate} = -\frac{\Delta[\text{C}_2\text{H}_4]}{\Delta t} = -\frac{\Delta[\text{O}_3]}{\Delta t} = +\frac{\Delta[\text{C}_2\text{H}_4\text{O}]}{\Delta t} = +\frac{\Delta[\text{O}_2]}{\Delta t}$$

$$\begin{aligned}\text{Rate} &= -\frac{\Delta[\text{O}_3]}{\Delta t} = -\frac{(1.10 \times 10^{-5} \text{ mol/L}) - (3.20 \times 10^{-5} \text{ mol/L})}{60.0 \text{ s} - 0.0 \text{ s}} \\ &= 3.50 \times 10^{-7} \text{ mol/L}\cdot\text{s}\end{aligned}$$

Table 16.1 Concentration of O_3 at Various Times in Its Reaction with C_2H_4 at 303 K

Time (s)	Concentration of O_3 (mol/L)
0.0	3.20×10^{-5}
10.0	2.42×10^{-5}
20.0	1.95×10^{-5}
30.0	1.63×10^{-5}
40.0	1.40×10^{-5}
50.0	1.23×10^{-5}
60.0	1.10×10^{-5}

This calculation gives us the **average rate** over that period; that is, during the first 60.0 s of the reaction, ozone concentration decreases an average of $3.50 \times 10^{-7} \text{ mol/L}$ each second.

Average rate of reaction

Between the starting time **0.0 s** and **10.0 s**, the average rate is

$$\begin{aligned}\text{Rate} &= -\frac{\Delta[\text{O}_3]}{\Delta t} = -\frac{(2.42 \times 10^{-5} \text{ mol/L}) - (3.20 \times 10^{-5} \text{ mol/L})}{10.0 \text{ s} - 0.0 \text{ s}} \\ &= 7.80 \times 10^{-7} \text{ mol/L}\cdot\text{s}\end{aligned}$$

During the last 10.0 s, between **50.0 s** and **60.0 s**, the average rate is

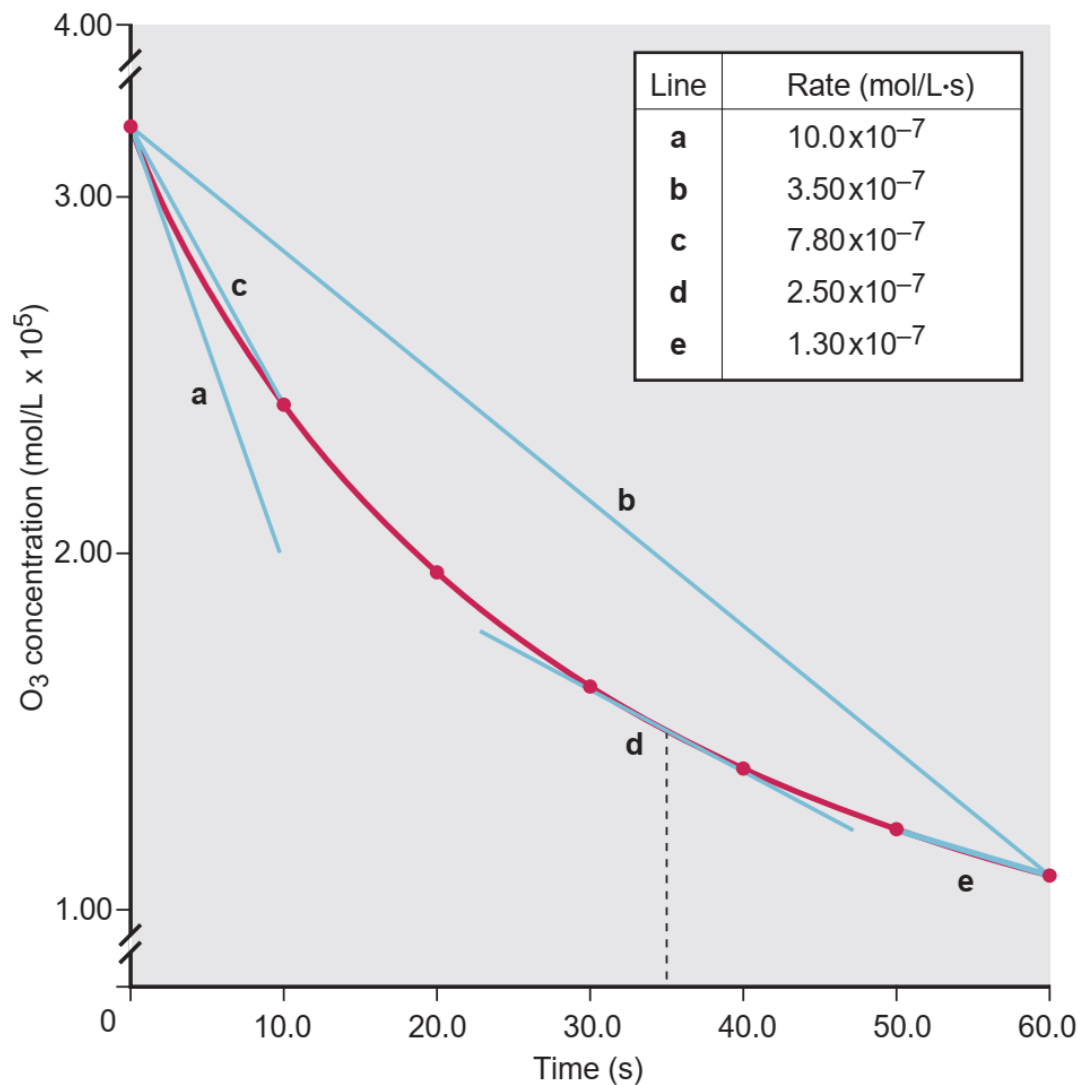
$$\begin{aligned}\text{Rate} &= -\frac{\Delta[\text{O}_3]}{\Delta t} = -\frac{(1.10 \times 10^{-5} \text{ mol/L}) - (1.23 \times 10^{-5} \text{ mol/L})}{60.0 \text{ s} - 50.0 \text{ s}} \\ &= 1.30 \times 10^{-7} \text{ mol/L}\cdot\text{s}\end{aligned}$$

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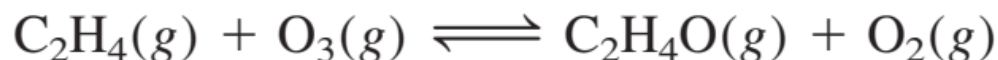
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Instantaneous Rate

The slope of a line tangent to the curve at a particular point gives the instantaneous rate at that time.



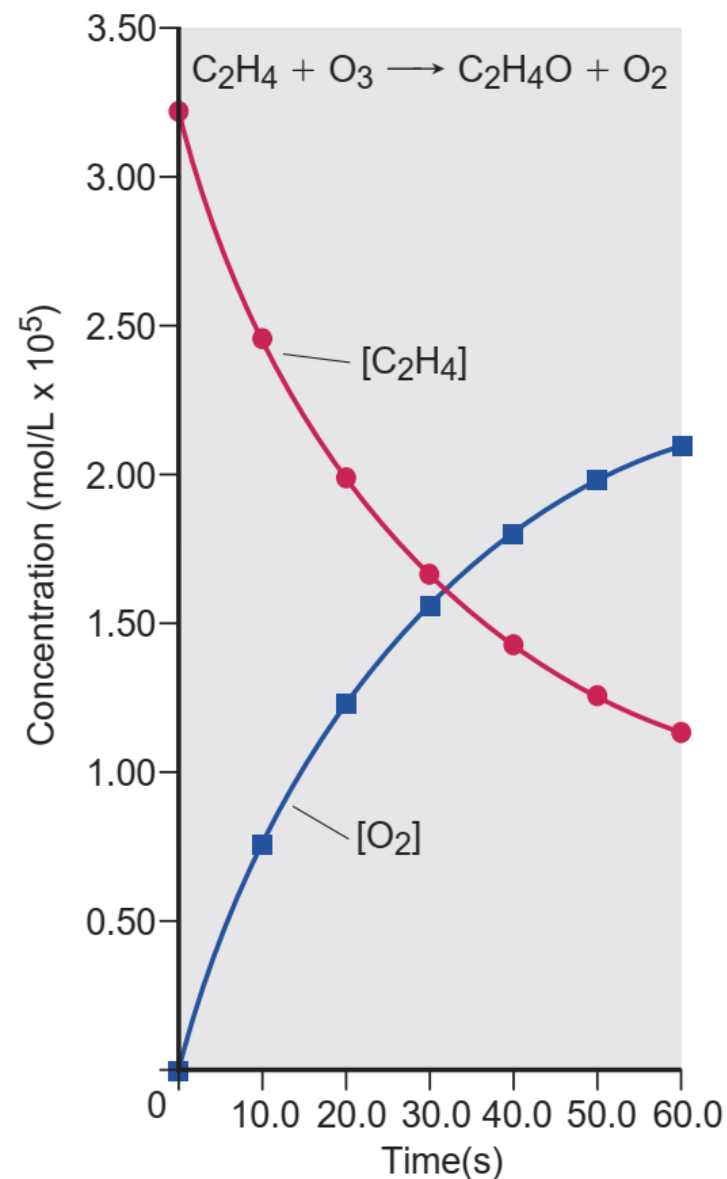
Rate of a reaction



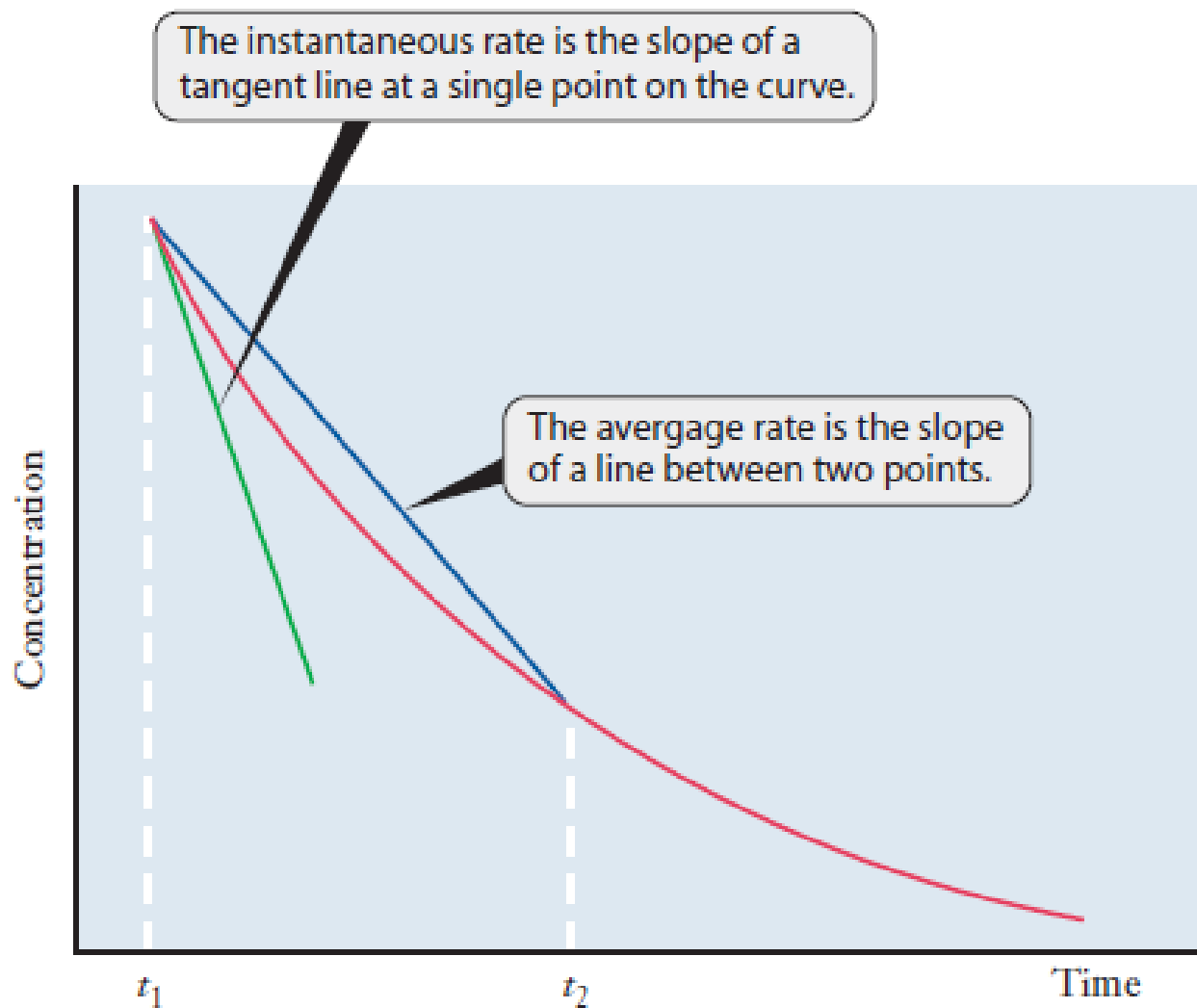
$$\text{Rate} = -\frac{\Delta[\text{C}_2\text{H}_4]}{\Delta t} = -\frac{\Delta[\text{O}_3]}{\Delta t} = +\frac{\Delta[\text{C}_2\text{H}_4\text{O}]}{\Delta t} = +\frac{\Delta[\text{O}_2]}{\Delta t}$$

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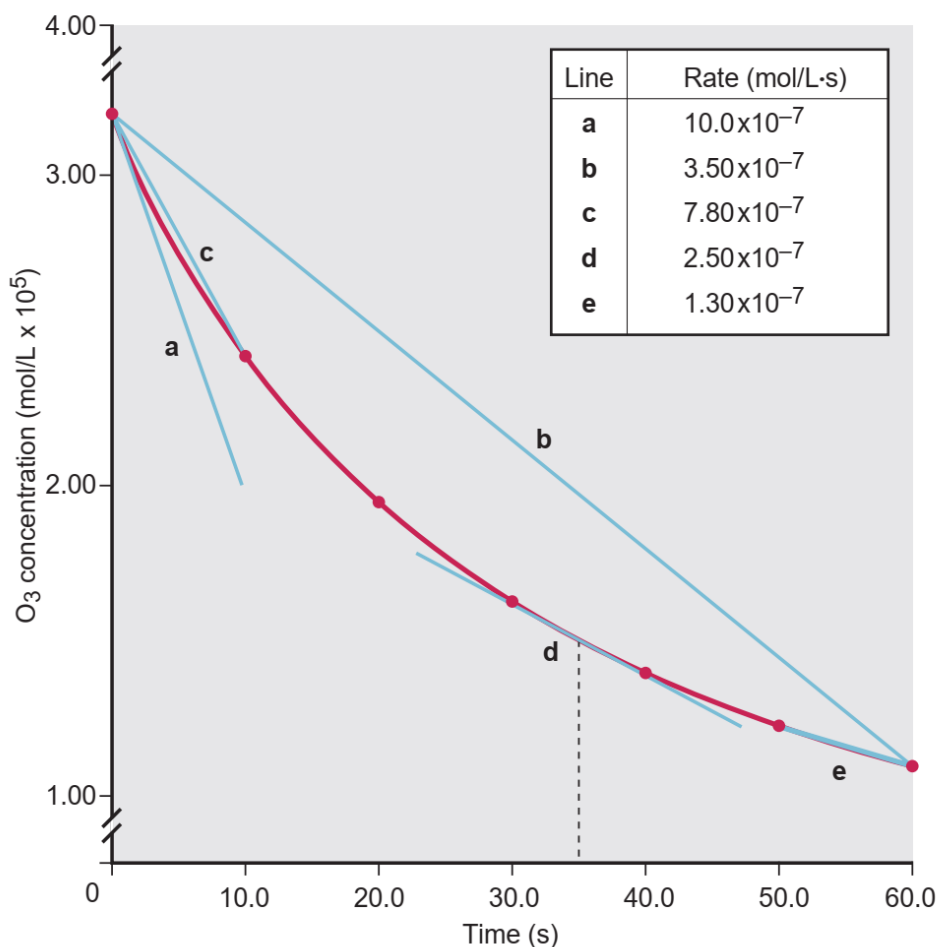
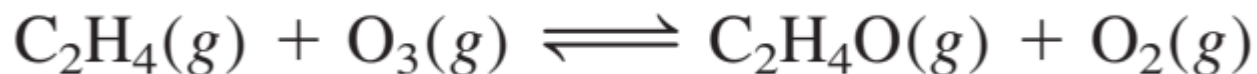
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50.0	1.23×10^{-5}
60.0	1.10×10^{-5}



Average Rate vs Instantaneous



Initial velocity (V_o)



The instantaneous rate measured at the beginning of the reaction is called *initial velocity of a reaction*. (e. g. slope of line a)

Initial velocity of disappearance of a reactant is higher than the velocities measured at later stages of reaction.

It is due to the presence of high concentration of reactants at time zero.

Rate and stoichiometry coefficients



The rate of the reaction can be written as:

$$\text{Rate} = -\frac{1}{a} \frac{\Delta[A]}{\Delta t} = -\frac{1}{b} \frac{\Delta[B]}{\Delta t} = \frac{1}{c} \frac{\Delta[C]}{\Delta t} = \frac{1}{d} \frac{\Delta[D]}{\Delta t}$$

QN

For the reaction **A + 3 B → 2 C**

How does the rate of disappearance of B compare to the rate of production of C?

- a. the rate of disappearance of B is 1/2 the rate of appearance of C
- b. the rate of disappearance of B is 3/2 the rate of appearance of C
- c. the rate of disappearance of B is 2/3 the rate of appearance of C
- d. the rate of disappearance of B is 1/3 the rate of appearance of C

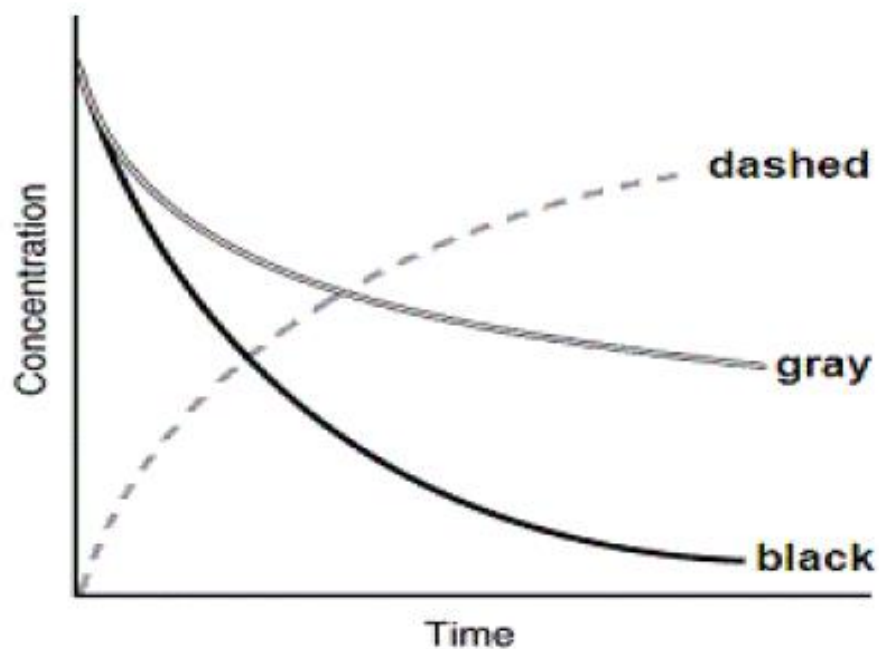
$$\frac{d[A]}{dt} = -\frac{1}{3} \frac{d[B]}{dt} = \frac{1}{2} \frac{d[C]}{dt}$$

Note: For relative rate comparison the rate must be always denoted as positive quantity

$$\frac{d[B]}{dt} = \frac{3}{2} \frac{d[C]}{dt}$$

QN

The following graph shows the kinetics curves for the reaction of oxygen with hydrogen to form water: $\text{O}_2(\text{g}) + 2\text{H}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$



Which curve correspond to the disappearance of hydrogen?

Problem:

Consider the reaction between oxygen and nitrogen monoxide, a key step in the formation of acid rain and in the industrial production of nitric acid:



If the rate of this reaction is $3.5 \times 10^{-3} \text{ mol/L.s}$, find out the rate of disappearance of $\text{O}_2 (\text{g})$ and $\text{NO} (\text{g})$, individually.

Solution:

$$\text{Rate} = -\frac{d[\text{O}_2]}{dt} = -\frac{1}{2} \frac{d[\text{NO}]}{dt}$$

Problem:

Reaction rate is expressed in terms of changes in concentration of reactants and products. Write a balanced equation for:

$$\text{Rate} = -\frac{1}{2} \frac{\Delta[\text{N}_2\text{O}_5]}{\Delta t} = \frac{1}{4} \frac{\Delta[\text{NO}_2]}{\Delta t} = \frac{\Delta[\text{O}_2]}{\Delta t}$$

The empirical Rate Law (or rate equation)



$$\text{Rate} = k[A]^m[B]^n \cdots$$

[A], [B]... Concentrations of A, B, etc.

k: Rate constant

m, n: Order of reaction

The actual values of the exponents *m* and *n* must be measured experimentally.

The rate law (or **rate equation**)

Let's consider a reaction: $A \longrightarrow B$.

$$Rate = k[A]^m$$

$[A] \rightarrow$ Concentration of A

$k \rightarrow$ Rate constant, is specific for a given reaction at a given temperature; it does *not* change as the reaction proceeds

$m \rightarrow$ Order of the reaction with respect to A.

Order define how the rate is affected by reactant concentration.

Significance of “Reaction Order”

For first order reaction:

In the simplest case, a reaction with a single reactant A, the reaction is *first order* overall if the rate is directly proportional to [A]:

$$\text{Rate} = k[\text{A}]$$

For second order reaction:

It is *second order* overall if the rate is directly proportional to the square of [A]:

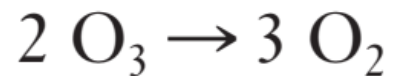
$$\text{Rate} = k[\text{A}]^2$$

For zero order reaction:

And it is *zero order* overall if the rate is *not* dependent on [A] at all, a situation that is quite common in metal-catalyzed and biochemical processes:

$$\text{Rate} = k[\text{A}]^0 = k(1) = k$$

What is the overall order for the following reaction?



$$\text{Rate} = k \frac{[\text{O}_3]^2}{[\text{O}_2]}$$

$$\text{Rate} = k \times [\text{O}_3]^2 \times [\text{O}_2]^{-1}$$

First order reaction in overall

Examples:

Here are some real examples. For the reaction between nitrogen monoxide and ozone,



the rate law has been experimentally determined to be

$$\text{Rate} = k[\text{NO}][\text{O}_3]$$

Now consider a different gas-phase reaction:



The rate law for this reaction has been determined to be

$$\text{Rate} = k[\text{NO}]^2[\text{H}_2]$$

Examples:

Finally, for the hydrolysis of 2-bromo-2-methylpropane, which we just considered in the Tools of the Laboratory essay,



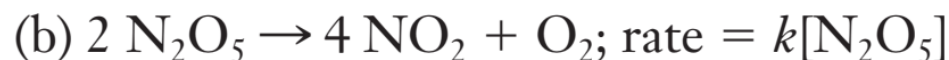
the rate law has been found to be

$$\text{Rate} = k[(\text{CH}_3)_3\text{CBr}]$$

This reaction is first order in 2-bromo-2-methylpropane. Note that the concentration of H_2O does not even appear in the rate law. Thus, the reaction is zero order with respect to H_2O ($[\text{H}_2\text{O}]^0$). This means that the rate does not depend on the concentration of H_2O . We can also write the rate law as

$$\text{Rate} = k[(\text{CH}_3)_3\text{CBr}][\text{H}_2\text{O}]^0$$

Check Your Understanding The following reactions were studied in the laboratory, and their respective rate laws were determined. Find the reaction orders with respect to each reactant and the overall order in each of the following:



Deriving the units of rate constants

Table 16.3 Units of the Rate Constant k for Several Overall Reaction Orders

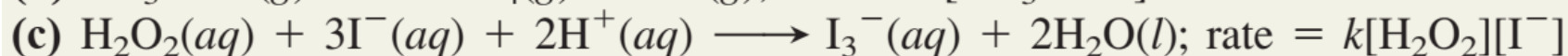
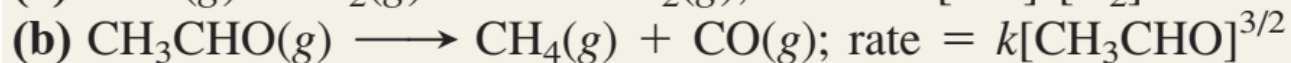
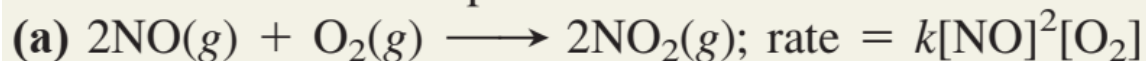
Overall Reaction Order	Units of k (t in seconds)
0	mol/L·s (or mol L ⁻¹ s ⁻¹)
1	1/s (or s ⁻¹)
2	L/mol·s (or L mol ⁻¹ s ⁻¹)
3	L ² /mol ² ·s (or L ² mol ⁻² s ⁻¹)

General formula:

$$\text{Units of } k = \frac{\left(\frac{\text{L}}{\text{mol}}\right)^{\text{order}-1}}{\text{unit of } t}$$

Problem:

PROBLEM For each of the following reactions, use the given rate law to determine the reaction order with respect to each reactant and the overall order:



Problem:

Give the individual reaction orders for all substances and the overall reaction order from the following rate law:

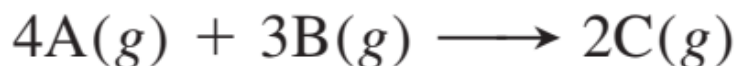
$$\text{Rate} = k \frac{[\text{O}_3]^2}{[\text{O}_2]}$$

By what factor does the rate change if each of the following changes occurs:

- (a) $[\text{O}_3]$ is doubled;
- (b) $[\text{O}_2]$ is doubled;
- (c) $[\text{O}_2]$ is halved?

Problem:

For the reaction



the following data were obtained at constant temperature:

Experiment	Initial [A] (mol/L)	Initial [B] (mol/L)	Initial Rate (mol/L·min)
1	0.100	0.100	5.00
2	0.300	0.100	45.0
3	0.100	0.200	10.0
4	0.300	0.200	90.0

(a) What is the order with respect to each reactant? (b) Write the rate law. (c) Calculate k (using the data from experiment 1).

Solution

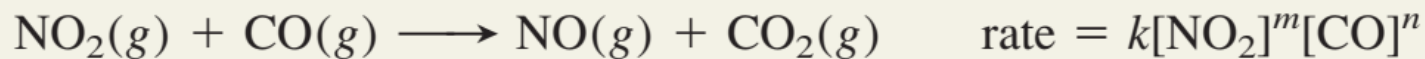
$$Rate = k[A]^2[B]^1$$

$$\text{Overall order} = 2 + 1 = 3$$

From experiment set-1, $5 = k \times (0.1)^2 \times (0.1)$

$$k = 5 \times 10^3 \text{ L}^2/\text{mol}^2.\text{min}^3$$

PROBLEM Many gaseous reactions occur in car engines and exhaust systems. One of these is



Use the following data to determine the individual and overall reaction orders:

Experiment	Initial Rate (mol/L·s)	Initial [NO ₂] (mol/L)	Initial [CO] (mol/L)
1	0.0050	0.10	0.10
2	0.080	0.40	0.10
3	0.0050	0.10	0.20

SOLUTION Calculating m in $[\text{NO}_2]^m$: We take the ratio of the rate laws for experiments 1 and 2, in which $[\text{NO}_2]$ varies but $[\text{CO}]$ is constant:

$$\frac{\text{Rate 2}}{\text{Rate 1}} = \frac{k[\text{NO}_2]_2^m[\text{CO}]_2^n}{k[\text{NO}_2]_1^m[\text{CO}]_1^n} = \left(\frac{[\text{NO}_2]_2}{[\text{NO}_2]_1}\right)^m \quad \text{or} \quad \frac{0.080 \text{ mol/L}\cdot\text{s}}{0.0050 \text{ mol/L}\cdot\text{s}} = \left(\frac{0.40 \text{ mol/L}}{0.10 \text{ mol/L}}\right)^m$$

This gives $16 = (4.0)^m$, so we have $m = \log 16 / \log 4.0 = 2.0$. The reaction is **second order in NO_2** .

Calculating n in $[\text{CO}]^n$: We take the ratio of the rate laws for experiments 1 and 3, in which $[\text{CO}]$ varies but $[\text{NO}_2]$ is constant:

$$\frac{\text{Rate 3}}{\text{Rate 1}} = \frac{k[\text{NO}_2]_3^2[\text{CO}]_3^n}{k[\text{NO}_2]_1^2[\text{CO}]_1^n} = \left(\frac{[\text{CO}]_3}{[\text{CO}]_1}\right)^n \quad \text{or} \quad \frac{0.0050 \text{ mol/L}\cdot\text{s}}{0.0050 \text{ mol/L}\cdot\text{s}} = \left(\frac{0.20 \text{ mol/L}}{0.10 \text{ mol/L}}\right)^n$$

We have $1.0 = (2.0)^n$, so $n = 0$. The rate does not change when $[\text{CO}]$ varies, so the reaction is **zero order in CO** .

Therefore, the rate law is

$$\text{Rate} = k[\text{NO}_2]^2[\text{CO}]^0 = k[\text{NO}_2]^2(1) = k[\text{NO}_2]^2$$

The reaction is **second order overall**.

Integral Rate Laws

$$\text{Rate} = k[A]^n$$

The rate law allows us to predict the rate of a reaction for a given set of concentrations.

However, the **rate law does not** let us easily *predict the concentrations or rate at some later time*.

To do that,

“we need to formulate an equation that tells us explicitly how the concentrations will change as a function of time”

This new equation, called the **integrated rate law**, can be derived from the rate law itself.

Integrated Rate Laws

Rate law



$$\text{Rate} = k [A]^m$$

Integrated rate laws allow us to calculate the concentration of reactants or products at any given time, during the course of a reaction.

The integrated rate law depends on the order of reaction

'0th' order

$$\text{Rate} = k [A]^0$$

'1st' order

$$\text{Rate} = k [A]^1$$

'2nd' order

$$\text{Rate} = k [A]^2$$

'nth' order

$$\text{Rate} = k [A]^n$$

Integrated Rate Laws

Zero Order Reactions



$$\text{Rate} = -\frac{d[A]}{dt} = k$$

at **t = 0 sec**, the Concentration of reactant = **[A]₀**

$$-d[A] = k \times dt$$

at **t = t sec**, the Concentration of reactant = **[A]_t**

$$-\int_{A_0}^{A_t} d[A] = \int_{t=0}^t k dt$$

$$[A]_0 - [A]_t = k t$$

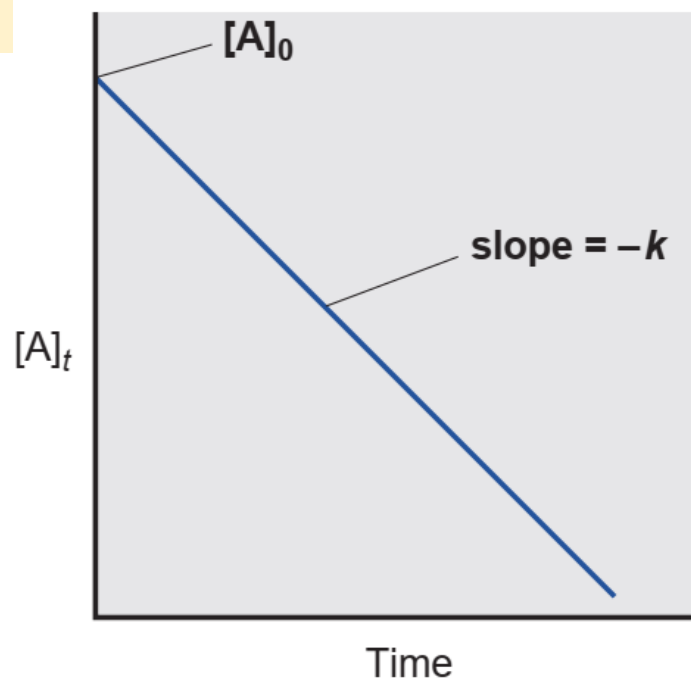
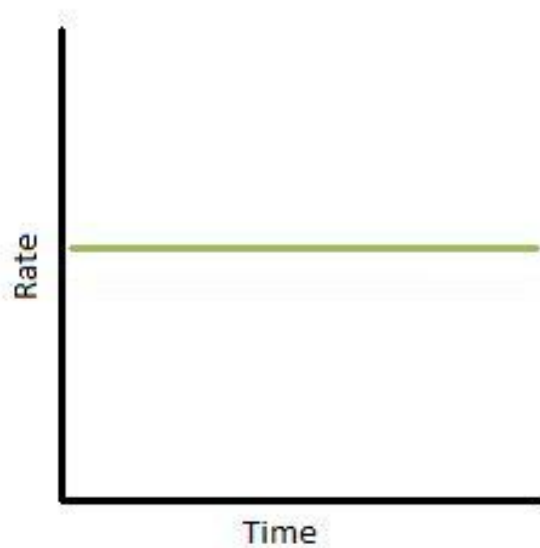
$$-([A]_t - [A]_0) = k (t - 0)$$

Zero Order Reactions

$$[A]_0 - [A]_t = k t$$

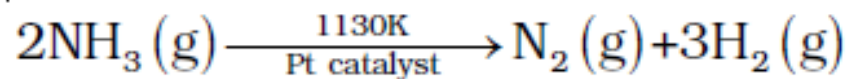
$$[A]_t = [A]_0 - k t$$

$$y = m x + C$$

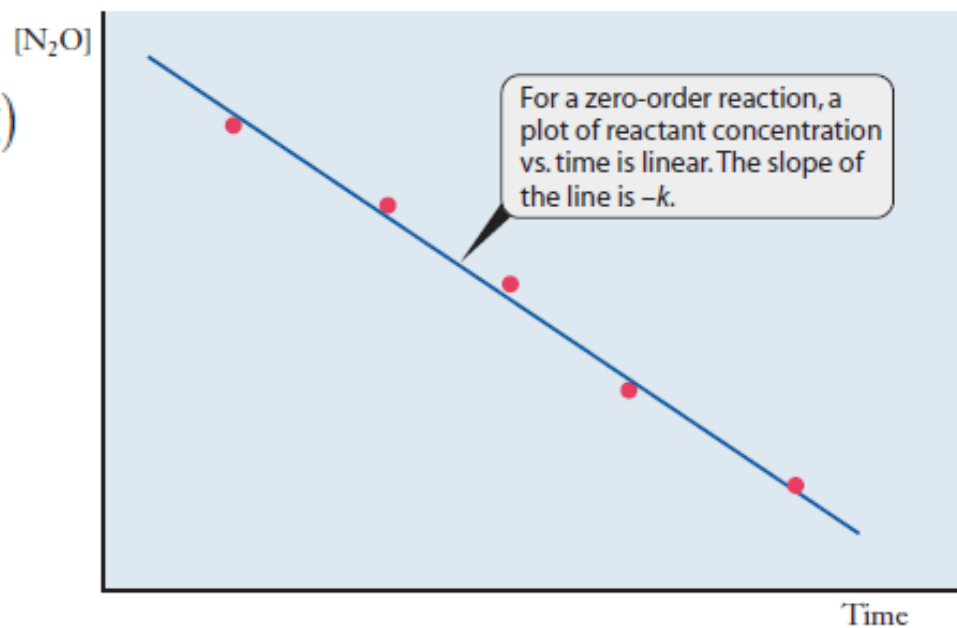


Zero Order Reactions

Ex: The decomposition of N_2O



$$\text{Rate} = k [\text{NH}_3]^0 = k$$



Reaction Half-Life ($t_{1/2}$)

The half-life of a reaction is the time in which the concentration of a reactant is reduced to one half of its initial concentration

When $t = t_{1/2}$ $[A]_{t_{1/2}} = \frac{[A]_0}{2}$

$$[A]_0 - [A]_t = k t$$

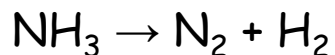
$$[A]_0 - \frac{[A]_0}{2} = k t_{1/2}$$

$$\frac{[A]_0}{2} = k t_{1/2}$$

$$t_{1/2} = \frac{[A]_0}{2 k}$$

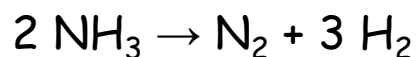
QN

Rate of decomposition of NH_3 on platinum surface is zero order reaction. What are the rate of products of H_2 and N_2 if $k = 2.5 \times 10^{-4} \text{ Ms}^{-1}$?

Ans.

$$\text{Rate} = k$$

$$\text{Rate} = 2.5 \times 10^{-4} \text{ Ms}^{-1}$$



$$\text{Rate} = -\frac{1}{2} \frac{d[\text{NH}_3]}{dt} = \frac{d[\text{N}_2]}{dt} = \frac{1}{3} \frac{d[\text{H}_2]}{dt}$$

$$\frac{d[\text{N}_2]}{dt} = 2.5 \times 10^{-4} \text{ Ms}^{-1}$$

$$\frac{d[\text{H}_2]}{dt} = 7.5 \times 10^{-4} \text{ Ms}^{-1}$$

For a zero order reaction, the rate constant is $0.14 \text{ mol L}^{-1} \text{ s}^{-1}$. If after two minutes the concentration of the reactant is 0.15 mol L^{-1} , calculate its initial concentration

Integrated Rate Laws

First Order Reactions



$$\text{rate} = k[A]$$

$$\text{Rate} = -\frac{d[A]}{dt} = k[A]$$

$$\ln ([A]_t - [A]_0) = -k t$$

$$-\frac{d[A]}{[A]} = k dt$$

$$\int_{[A]_0}^{[A]_t} -\frac{d[A]}{[A]} = \int_0^t k dt$$

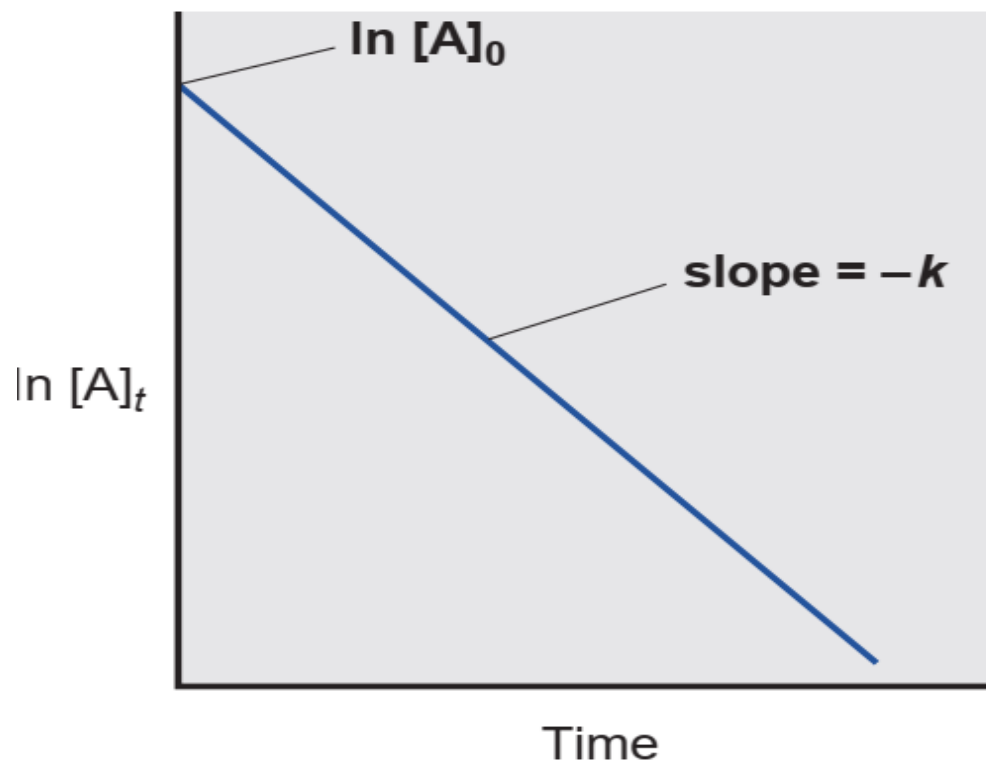
$$\ln \frac{[A]_0}{[A]_t} = kt$$

First Order Reactions

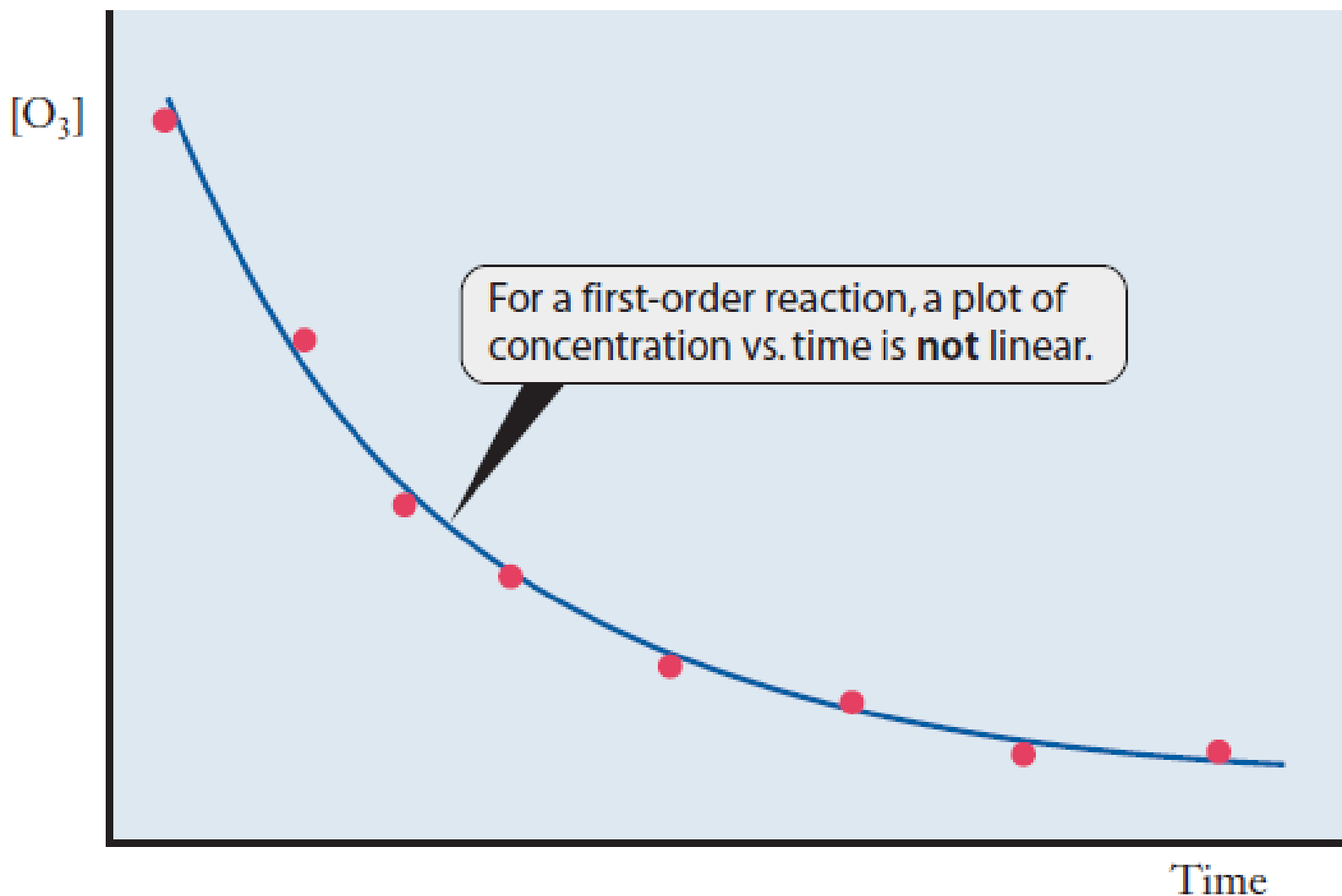
$$\ln \frac{[A]_0}{[A]_t} = kt$$

$$k \times t = \ln [A]_0 - \ln [A]_t$$

$$\ln [A]_t = \ln [A]_0 - k \times t$$

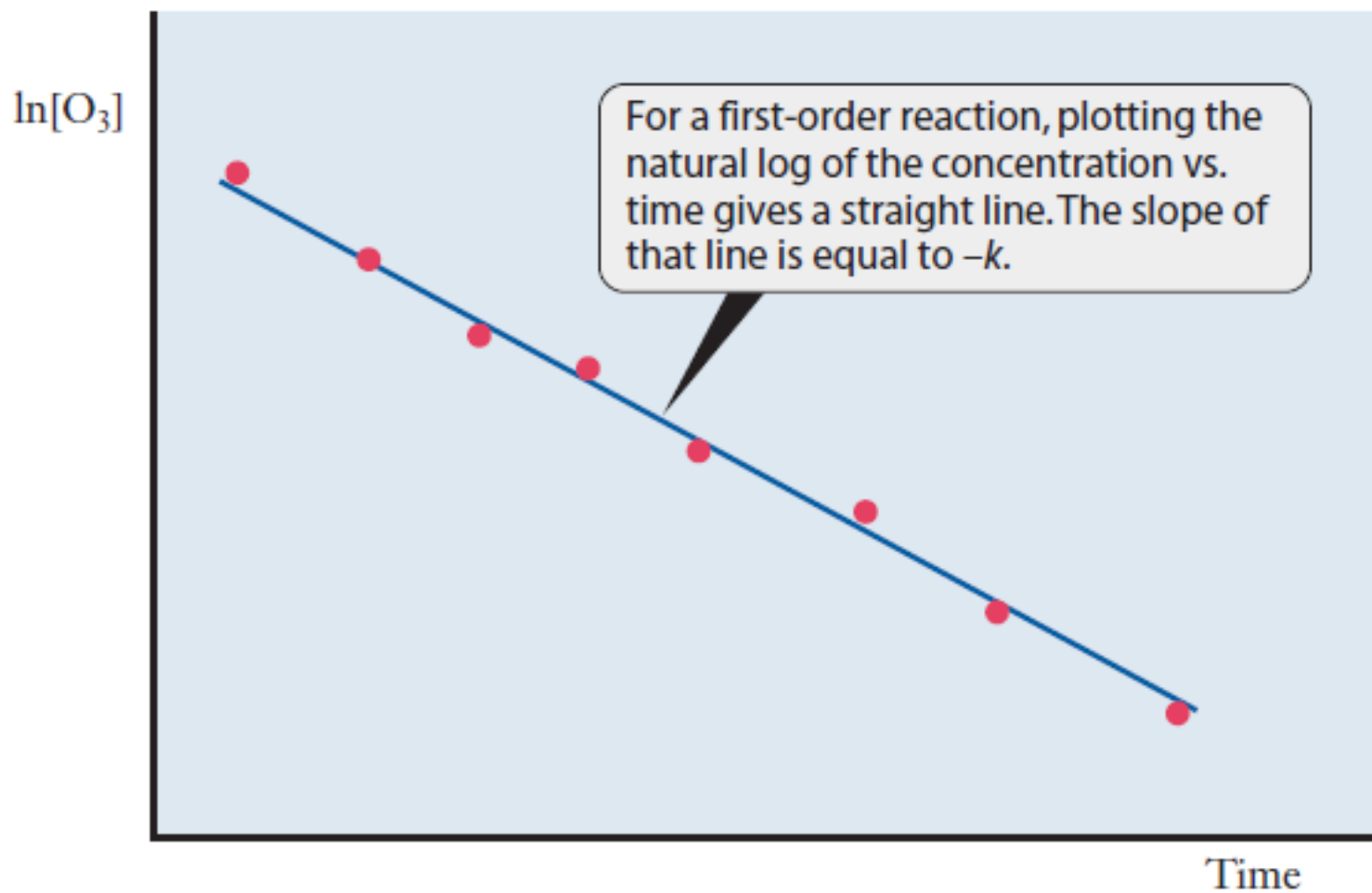


Dissociation O_3 follows first order kinetics



The linear form of integral rate expression for the first order reaction is $\ln[A] = -kt + \ln[A]_0$

Hence, the plot of $\ln[O_3]$ Vs t must be linear with a slope of $-k$



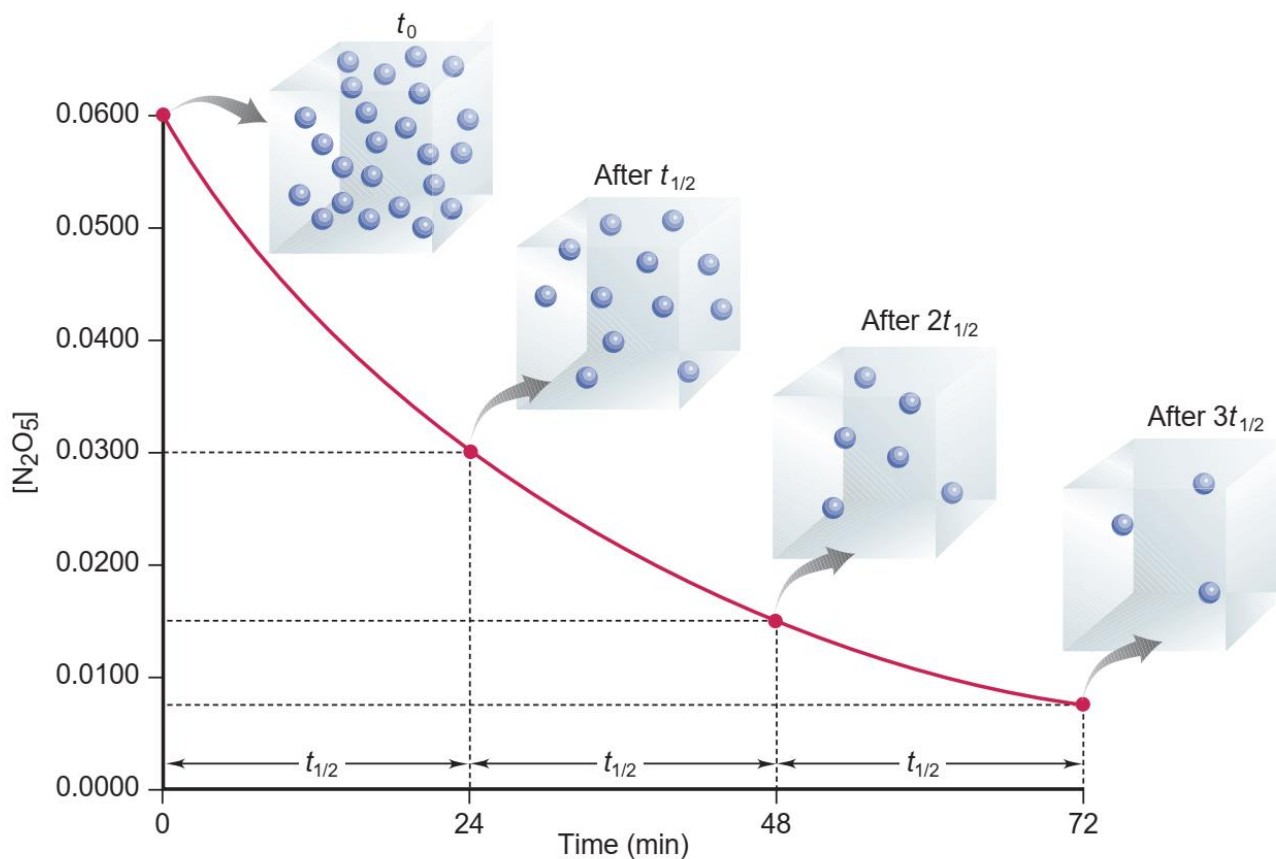
First Order Reactions

$$k \times t = 2.303 \log \left(\frac{[A]_0}{[A]_t} \right)$$

$$k \ t = 2.303 \log \left(\frac{a}{a-x} \right)$$

Reaction Half-life

The **half-life** ($t_{1/2}$) of a reaction is the time required for the reactant concentration to reach half its initial value.



For first order reaction:

$$\ln \frac{[A]_0}{[A]_t} = kt$$

After one half-life, $t = t_{1/2}$, and $[A]_t = \frac{1}{2}[A]_0$. Substituting, we obtain

$$\ln \frac{\cancel{[A]_0}}{\frac{1}{2}\cancel{[A]_0}} = kt_{1/2} \quad \text{or} \quad \ln 2 = kt_{1/2}$$

Then, solving for $t_{1/2}$, we have

$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k} \quad (\text{first-order process; rate} = k[A])$$

Half-Life

“The half-life of a reactant is the time takes for its concentration to fall to one-half of its original value”

When a first order reaction proceeding for one half-life, the concentration of the Substance must be $[A] = [A]_0/2$.

Thus we have,

$$\ln \left(\frac{\frac{1}{2}[A]_0}{[A]_0} \right) = \ln \left(\frac{[A]_0}{2[A]_0} \right) = -kt_{1/2}$$

$$\ln \frac{1}{2} = -kt_{1/2}$$

$$\ln 1 - \ln 2 = -kt_{1/2}$$

$$\ln 2 = kt_{1/2}$$

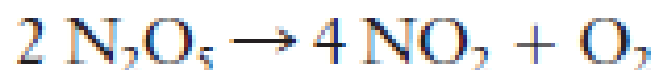
$$t_{1/2} = \frac{\ln 2}{k} = \frac{0.693}{k}$$

Example 1: The initial concentration of N_2O_5 in the following first order reaction



was $1.24 \times 10^{-2} \text{ mol L}^{-1}$ at 318 K. The concentration of N_2O_5 after 60 minutes was $0.20 \times 10^{-2} \text{ mol L}^{-1}$. Calculate the rate constant of the reaction at 318 K.

11.39 The decomposition of N_2O_5 in solution in carbon tetrachloride is a first-order reaction.



The rate constant at a given temperature is found to be $5.25 \times 10^{-4} \text{ s}^{-1}$. If the initial concentration of N_2O_5 is 0.200 M , what is its concentration after exactly 10 minutes have passed?

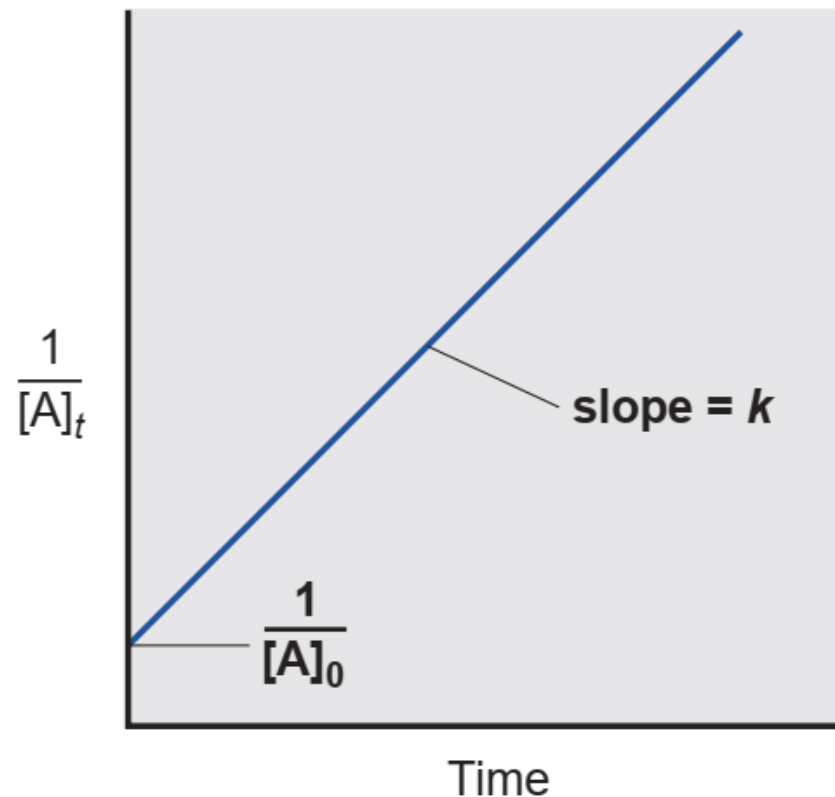
11.40 In Exercise 11.39 above, if the initial concentration of N_2O_5 is 0.100 M, how long will it take for the concentration to drop to 0.0100 times its original value?

Integrated Rate Law for Second Order Reactions



(second-order reaction; rate = $k[A]^2$)

$$\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$$



Summary of Rate Law, Rate Constants, Half-Lives

	Zero Order	First Order	Second Order
Differential rate law	$\text{Rate} = k$	$\text{Rate} = k[A]$	$\text{Rate} = k[A]^2$
Integrated rate law	$[A]_t = [A]_0 - kt$	$\ln[A]_t = \ln[A]_0 - kt$	$1/[A]_t = 1/[A]_0 + kt$
Linear plot	$[A]_t$ vs t	$\ln[A]_t$ vs t	$1/[A]_t$ vs t
Rate constant units	$M \cdot s^{-1}$	s^{-1}	$M^{-1} \cdot s^{-1}$
Half life	$\frac{[A]_0}{2k}$	$\frac{0.693}{k}$	$1/k[A]_0$

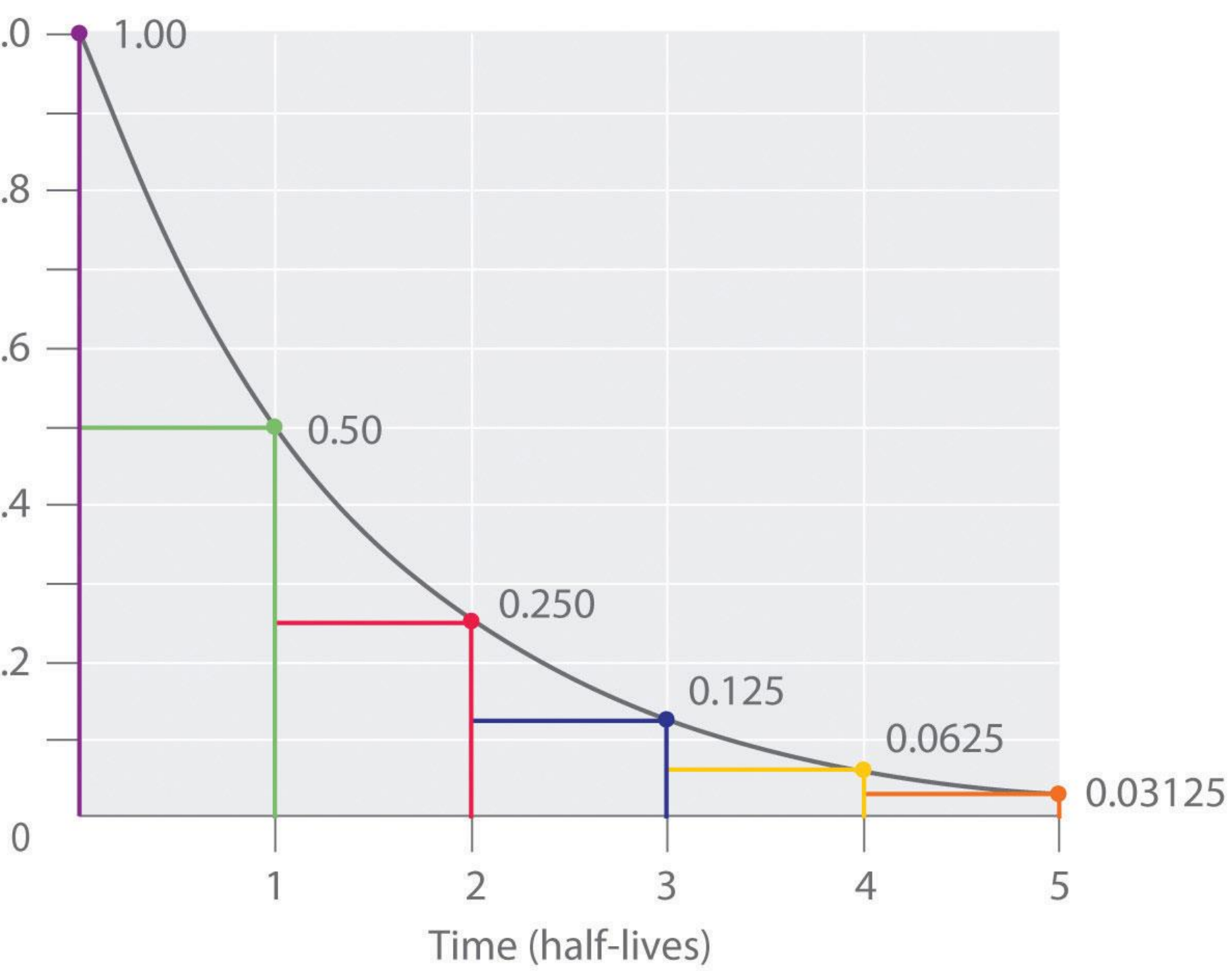
Unit for Rate = $\text{Mol L}^{-1} \text{s}^{-1} = M \text{s}^{-1}$

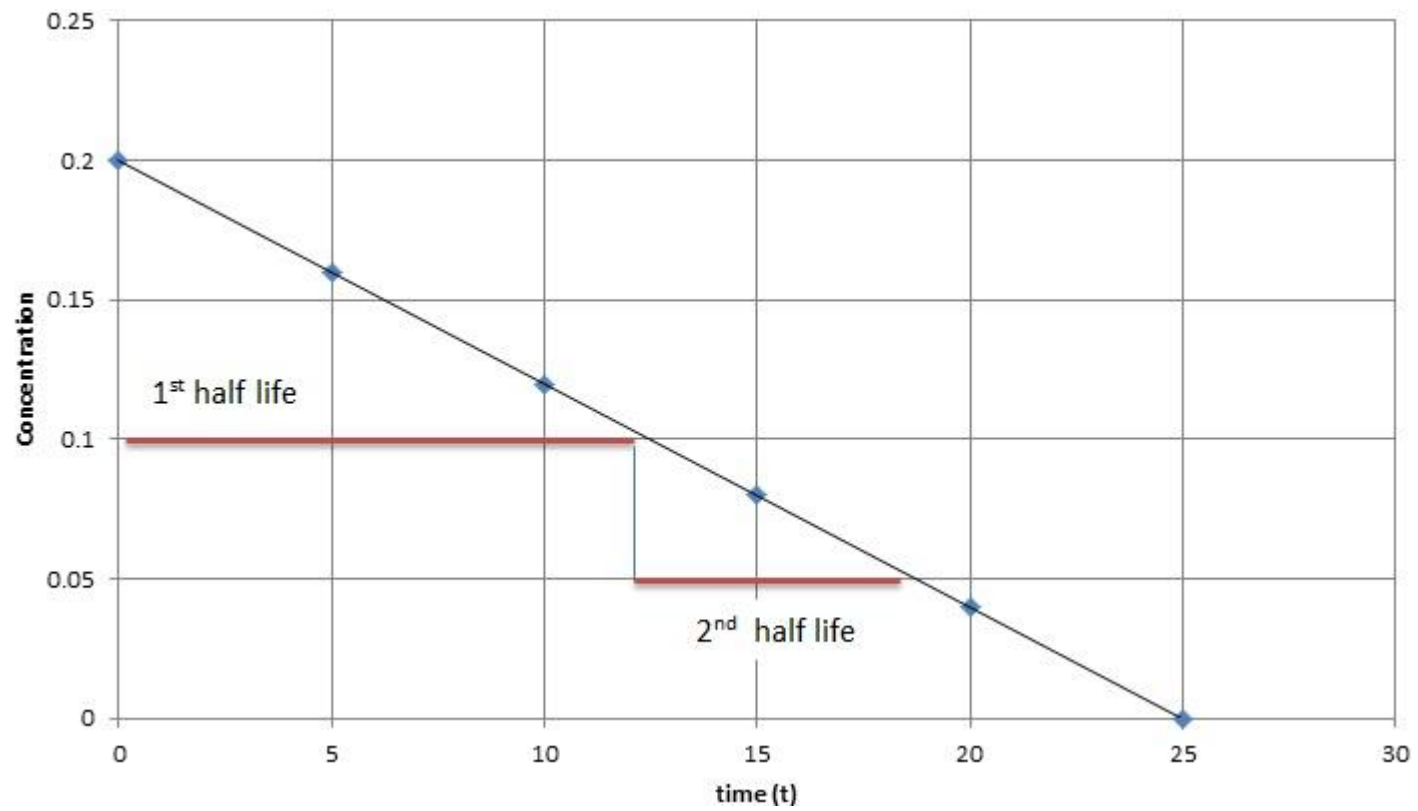
Rate = $k[A]$

$k = s^{-1}$

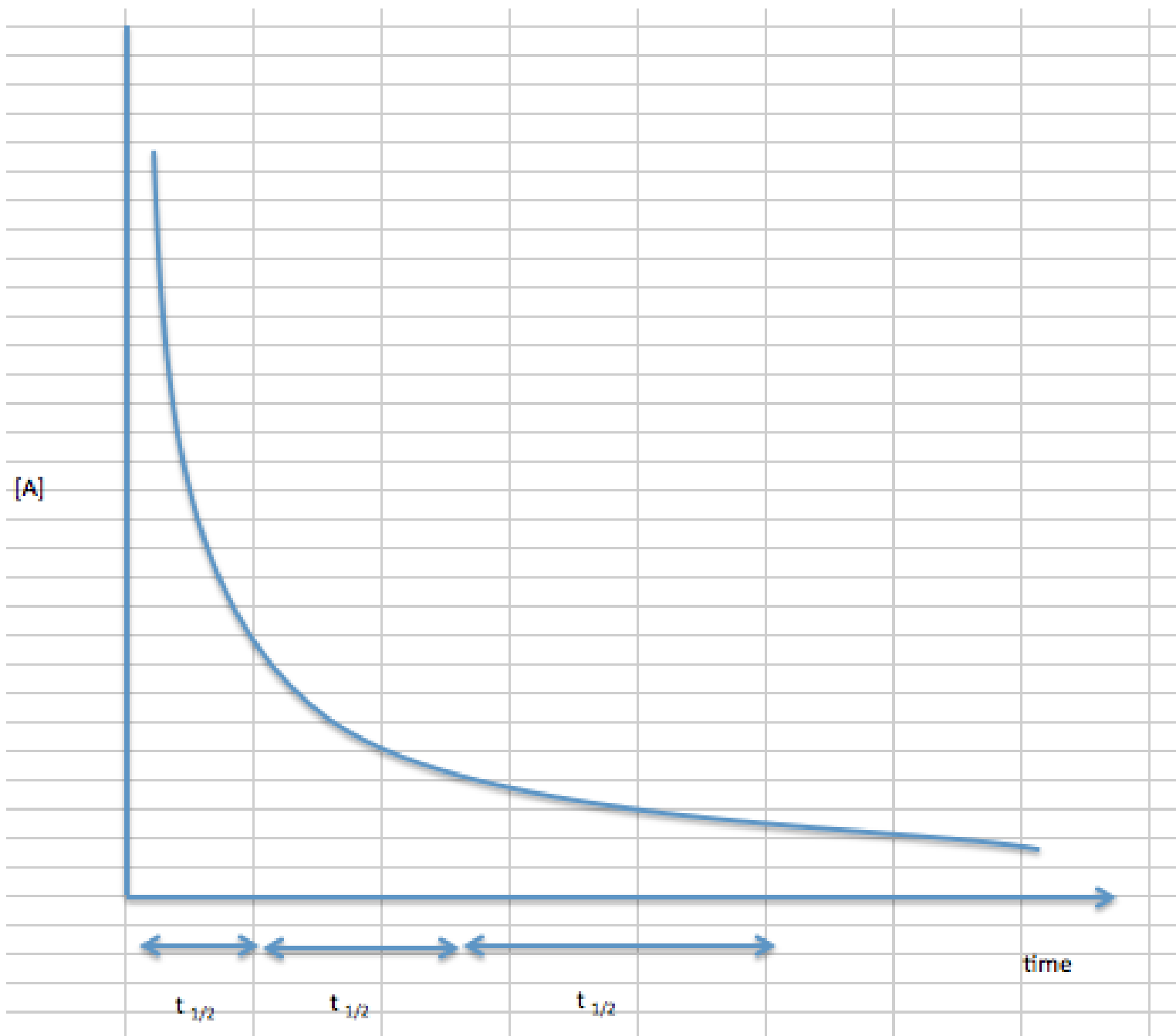
Unit for concentration = $\text{Mol L}^{-1} = M$

$k = \text{Rate}/[A] = M \text{s}^{-1} / M$





M



Reaction Mechanism

A **reaction mechanism** is a collection of one or more molecular steps that account for the way reactants become products



Elementary Reactions and **Molecularity**

The individual steps, which together make up a proposed reaction mechanism, are called **elementary reactions**

An elementary step is characterized by its **molecularity**, the **number of reactant** particles involved in the step.

To understand reaction mechanisms, we need to distinguish between the overall stoichiometry of a reaction and the steps in the reaction mechanism.



Summary of molecularity of elementary reactions

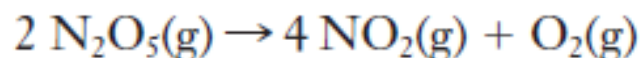
Type of Elementary Reaction	Molecularity	Rate Law
$A \rightarrow \text{products}$	Unimolecular	$\text{Rate} = k[A]$
$A + B \rightarrow \text{products}$	Bimolecular	$\text{Rate} = k[A][B]$
$2 A \rightarrow \text{products}$		$\text{Rate} = k[A]^2$
$A + B + C \rightarrow \text{products}$	Termolecular	$\text{Rate} = k[A][B][C]$
$2 A + B \rightarrow \text{products}$		$\text{Rate} = k[A]^2[B]$
$A + B + C + D \rightarrow \text{products}$	Not observed	



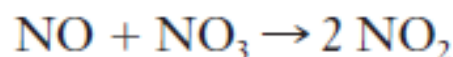
- The **ClO** generated in the first reaction is consumed in the second reaction.
- A chemical species that is generated in one step and consumed in a later step is called an intermediate or a **reactive intermediate**.
- Many mechanisms involve one or more intermediates.

- When the steps of the mechanism are properly summed, they give the observed stoichiometry of the overall reaction.

The decomposition of N_2O_5 is given by the equation



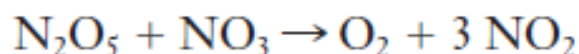
The following mechanism is proposed for this reaction:



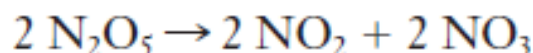
(a) Does this mechanism as written provide the correct stoichiometry? If not how does it need adjustment? (b) Identify all intermediates in the mechanism. (c) Identify the molecularity of each step in the mechanism.

Solution

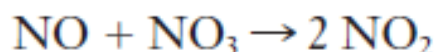
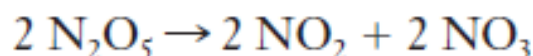
(a) A simple sum of the mechanism yields



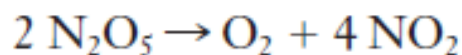
This is not the desired stoichiometry, but if we multiply the first step by 2, we have



Adding that to steps 2 and 3,



yields

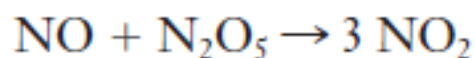
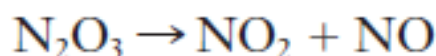


So the mechanism *does* provide the correct stoichiometry, as long as we realize that some steps may need to occur more often than others.

(b) The intermediates in this case are NO_3 and NO .

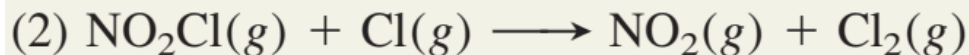
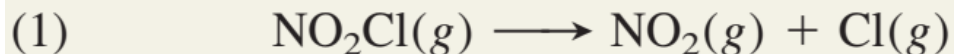
(c) The first step is unimolecular, whereas steps two and three are bimolecular.

Check Your Understanding The following mechanism was once proposed for the decomposition of N_2O_5 :



- (a) Is this mechanism plausible? (b) Identify the intermediates in the reaction.
(c) Determine the molecularity of each step.
-

PROBLEM The following two reactions are proposed as elementary steps in the mechanism for an overall reaction:

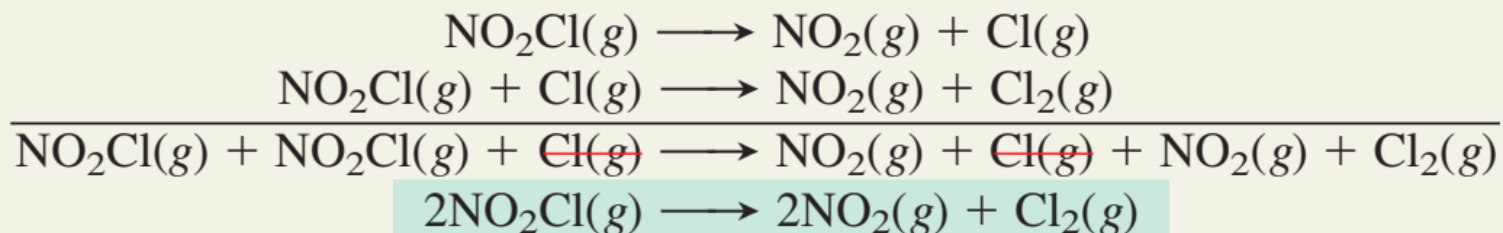


(a) Write the overall balanced equation.

(b) Determine the molecularity of each step.

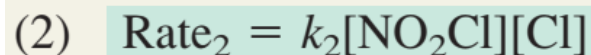
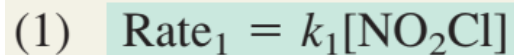
(c) Write the rate law for each step.

SOLUTION (a) Writing the overall balanced equation:



(b) Determining the molecularity of each step: The first elementary step has only one reactant, NO_2Cl , so it is **unimolecular**. The second elementary step has two reactants, NO_2Cl and Cl , so it is **bimolecular**.

(c) Writing rate laws for the elementary reactions:



Problem:



has an E_a of 1.00×10^2 kJ/mol and a rate constant of 0.286 L/mol·s at 500. K. What is the rate constant at 490. K?

$$\begin{aligned} \ln \frac{0.286 \text{ L/mol}\cdot\text{s}}{k_1} &= -\frac{1.00 \times 10^5 \text{ J/mol}}{8.314 \text{ J/mol}\cdot\text{K}} \\ &\quad \times \left(\frac{1}{500. \text{ K}} - \frac{1}{490. \text{ K}} \right) \\ &= 0.491 \\ k_1 &= 0.175 \text{ L/mol}\cdot\text{s} \end{aligned}$$

Effect of temperature on reaction rate

Have you ever asked yourself why you need a refrigerator? what advantages does the cooled storage space provide?

The answer lies in the **temperature dependence** of the rates of chemical reactions. Without a refrigerator, food would spoil more quickly.

Set up one sample of milk at room temperature and another sample in the refrigerator. Determine the time it takes the milk to smell sour. It is pretty obvious how this will turn out: the room temperature milk will sour much faster.



Effect of temperature on reaction rate

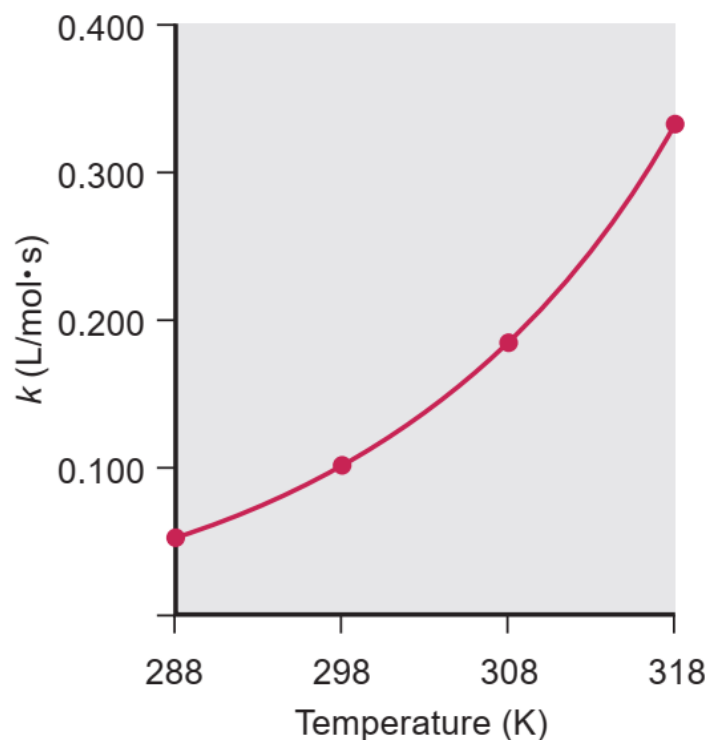


Figure 16.10 Dependence of the rate constant on temperature. **A**, In the hydrolysis of the ester ethyl acetate, $\text{CH}_3\text{COOCH}_2\text{CH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{CH}_3\text{CH}_2\text{OH}$ when reactant concentrations are held constant and temperature increases, the rate and rate constant increase. Note the near doubling of k with each rise of 10 K (10°C). **B**, A plot of rate constant vs. temperature for this reaction shows an exponentially increasing curve.

Expt	[Ester]	[H ₂ O]	<i>T</i> (K)	Rate (mol/L·s)	<i>k</i> (L/mol·s)
1	0.100	0.200	288	1.04×10^{-3}	0.0521
2	0.100	0.200	298	2.02×10^{-3}	0.101
3	0.100	0.200	308	3.68×10^{-3}	0.184
4	0.100	0.200	318	6.64×10^{-3}	0.332

Temperature Coefficient

- The ratio of the rate constants of a reaction at two temperatures differing by 10°C is known as temperature coefficient of the reaction.

$$\text{Temperature coefficient} = \frac{\text{Rate constant at } 350\text{C}}{\text{Rate constant at } 250\text{C}} = \frac{k_{35}}{k_{25}}$$

Temperature coefficient of a reaction is '2'. When temperature is increased from 30 to 90, the rate of reaction is increased by :

- (a) 32 (b) 64 (c) 60 (d) 128

The temperature coefficient of a reaction is 2, by what factor the rate of reaction increases when temperature is increased from 30°C to 80°C.

- (a) 32 (b) 64 (c) 60 (d) 128

For every 10°C rise in temperature, rate of reaction increased by 2 times. Then what happens to the rate if the temperature increased from 60°C to 80°C

Effect of temperature on reaction rate

Arrhenius equation

$$k = Ae^{-E_a/RT}$$

The E_a term is the **activation energy** of the reaction, which Arrhenius considered the *minimum energy* the molecules must have to react

A is a proportionality constant called the **frequency factor** or **pre-exponential factor**, which is related to the orientation of the colliding molecules.

Role of Activation energy (E_a)

Activation energy (E_a) of the reaction is considered as the *minimum energy* the molecules must have to react.

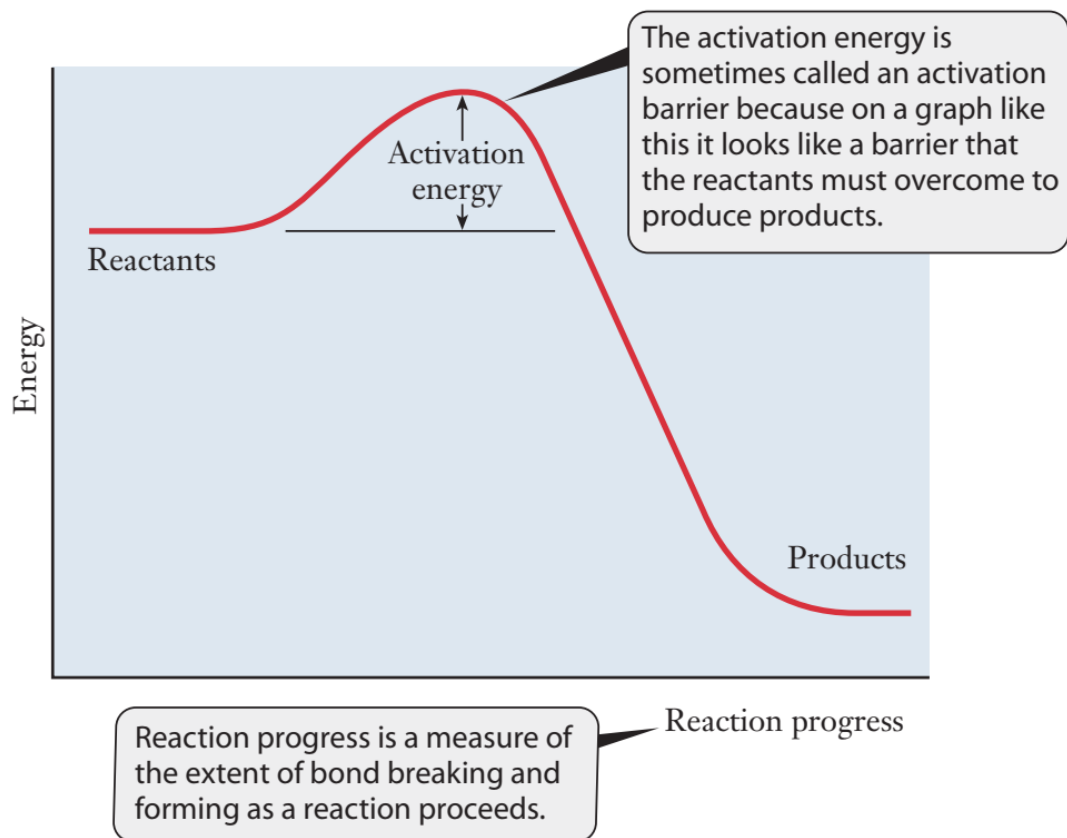
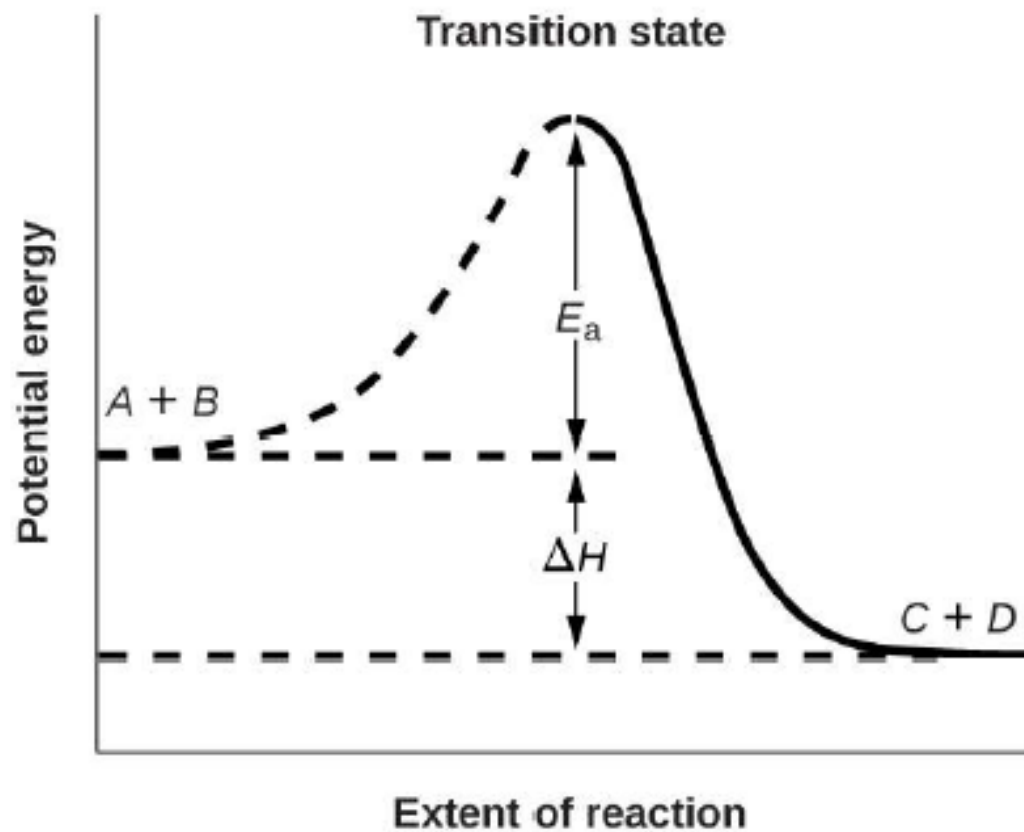


Figure 11.11 ■ Potential energy plot for an exothermic chemical reaction. To progress from reactants to products, the molecules must collide with enough energy to pass over the activation barrier.

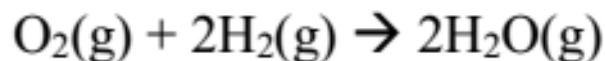
Activation Energy



Activation Energy

-A thermodynamically allowed reaction may be kinetically forbidden

The reaction that occurs is the combustion of H₂ as seen in the reaction below.



$\Delta H = -\#$ (combustion reaction)

$\Delta S = -\#$ (3 moles of gas turns into 2 moles of gas, therefore, positional probability decreases)

$$\Delta G^\circ = \left(-484 \frac{\text{kJ}}{\text{mol}}\right) - (298\text{K})\left(-0.089 \frac{\text{kJ}}{\text{molK}}\right) = -457 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta H_{\text{rxn}}^\circ = -484 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta S_{\text{rxn}}^\circ = -89 \frac{\text{J}}{\text{molK}}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

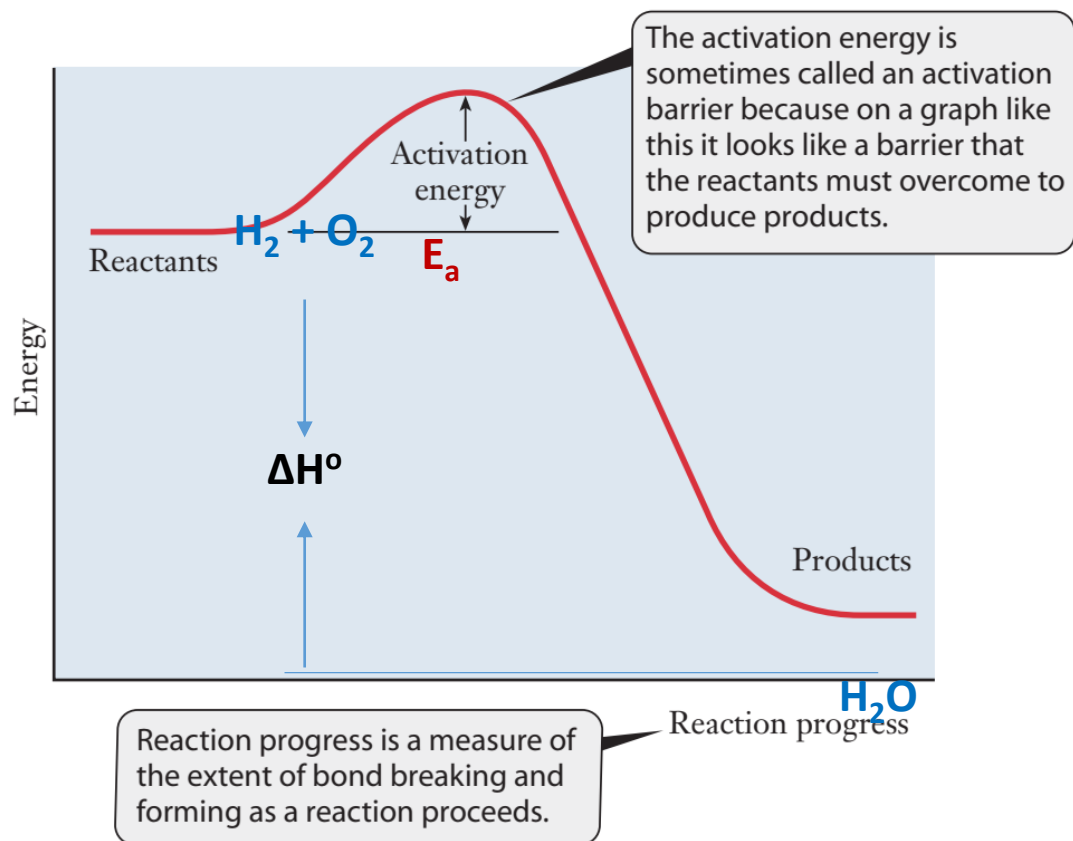
According to thermodynamics a negative ΔG indicates that a reaction is spontaneous.

However, *this reaction does not occur under ambient (normal) condition* due the kinetic reason. It needs some *energy input to activate the reaction*.

Role of Activation energy (E_a)

Activation energy (E_a) of the reaction is considered as the *minimum energy* the molecules must have to react.

Normally the activation energy is supplied by heat (Thermal energy)



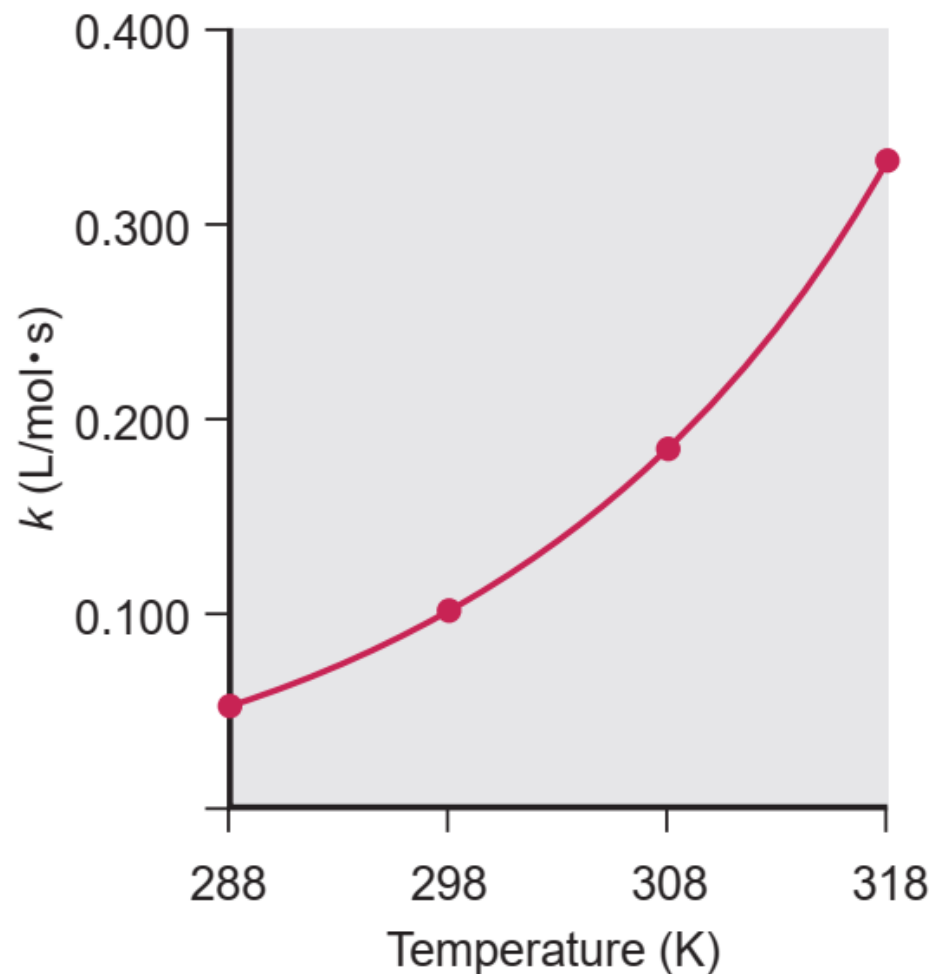
$$E_a(\text{backward}) = E_a(\text{forward}) + \Delta H$$

Effect of temperature on reaction rate

$$k = Ae^{-E_a/RT}$$

This negative exponential relationship between T and k means that - *as T increases, the negative exponent becomes smaller, so the value of k becomes larger, which means that the rate increases:*

Higher $T \Rightarrow$ larger k :
 $\Rightarrow$ increased rate

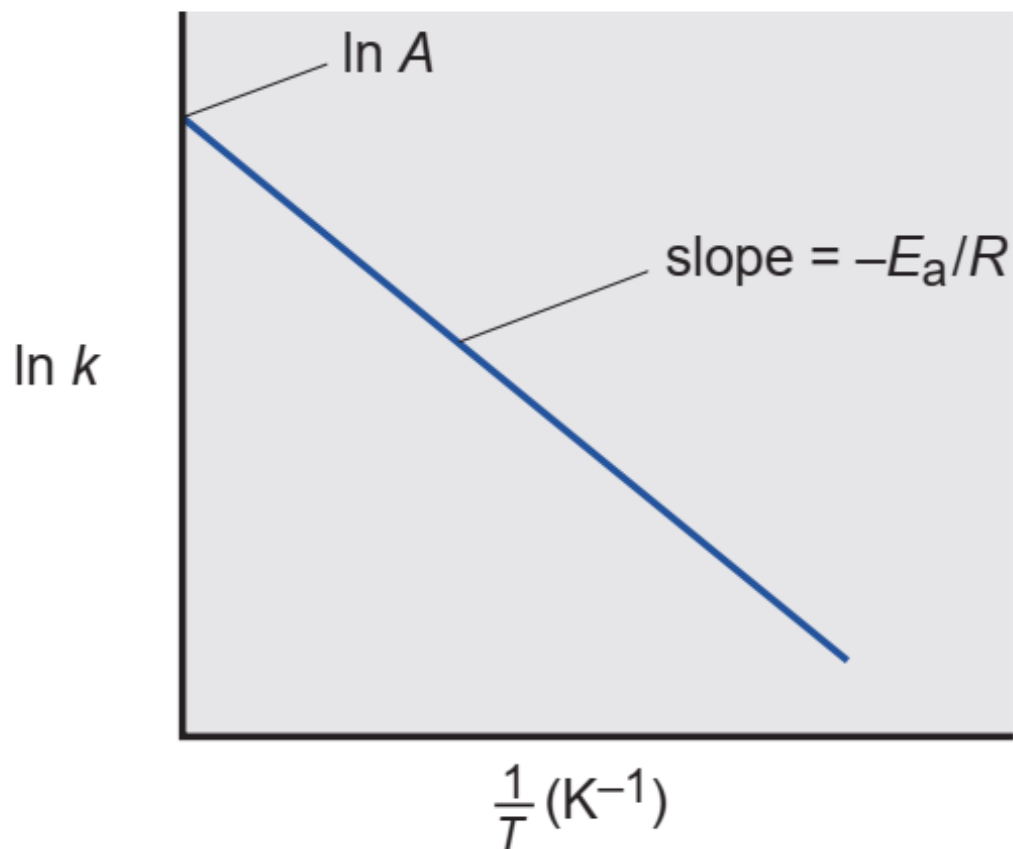


$$k = Ae^{-E_a/RT}$$

$$\ln k = \ln A - \frac{E_a}{R} \left(\frac{1}{T} \right)$$

$$y = b + mx$$

$$\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



R: Universal gas constant, 8.314 J/mol.K

Determination of E_a From Experimental Measurements

Method 2

$$\ln \frac{k_2}{k_1} = -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

R: Universal gas constant,
8.314 J/mol.K

PROBLEM The decomposition of hydrogen iodide,



has rate constants of $9.51 \times 10^{-9} \text{ L/mol}\cdot\text{s}$ at 500. K and $1.10 \times 10^{-5} \text{ L/mol}\cdot\text{s}$ at 600. K. Find E_a .

$$\begin{aligned}\ln \frac{k_2}{k_1} &= -\frac{E_a}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \\ E_a &= -R \left(\ln \frac{k_2}{k_1} \right) \left(\frac{1}{T_2} - \frac{1}{T_1} \right)^{-1} \\ &= -(8.314 \text{ J/mol}\cdot\text{K}) \left(\ln \frac{1.10 \times 10^{-5} \text{ L/mol}\cdot\text{s}}{9.51 \times 10^{-9} \text{ L/mol}\cdot\text{s}} \right) \left(\frac{1}{600. \text{ K}} - \frac{1}{500. \text{ K}} \right)^{-1} \\ &= 1.76 \times 10^5 \text{ J/mol} \\ &= 1.76 \times 10^2 \text{ kJ/mol}\end{aligned}$$

Collision Theory

- The dependence of rate of a reaction (or rate constant) on thermal energy was first model by Svante **Arrhenius** (1889).
- He combined the kinetic theories of gas with Boltzmann distribution law of molecular speeds to develop his mathematical formalism that *relates the rate constant with activation energy (E_a) and Temperature (T)*.

Collision theory model was originally developed for *gas phase reactions*.

Based on the following assumptions

1. Colliding molecules are *hard spheres*
2. In order for molecules to react, *they must collide* with each other
3. These collisions must also be *energetic enough* to be able to break and reform molecular bonds
4. The reactant atoms must be *oriented* in such a way that they can easily rearrange to form the products.

Maxwell-Boltzmann Distribution of Molecular Speeds

“In order for the reaction to happen, colliding molecules must possess certain minimum velocity (or kinetic energy)”

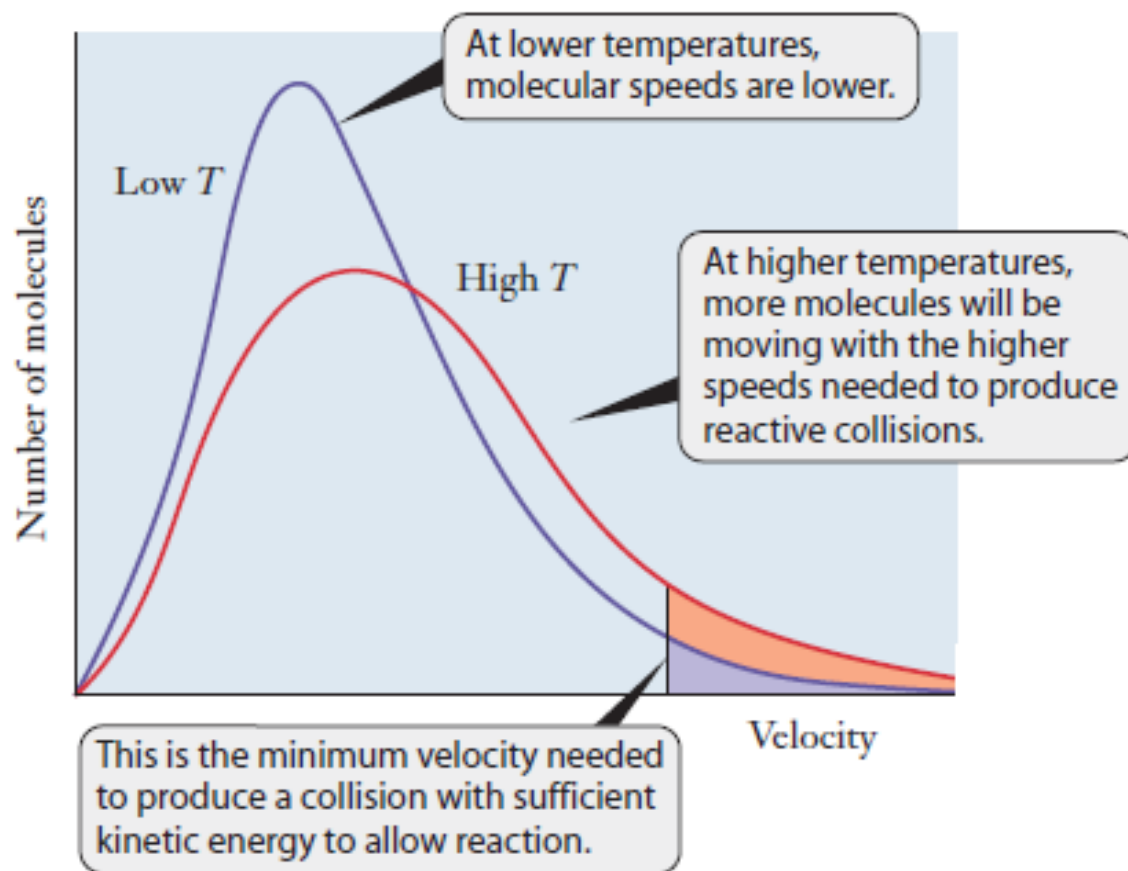
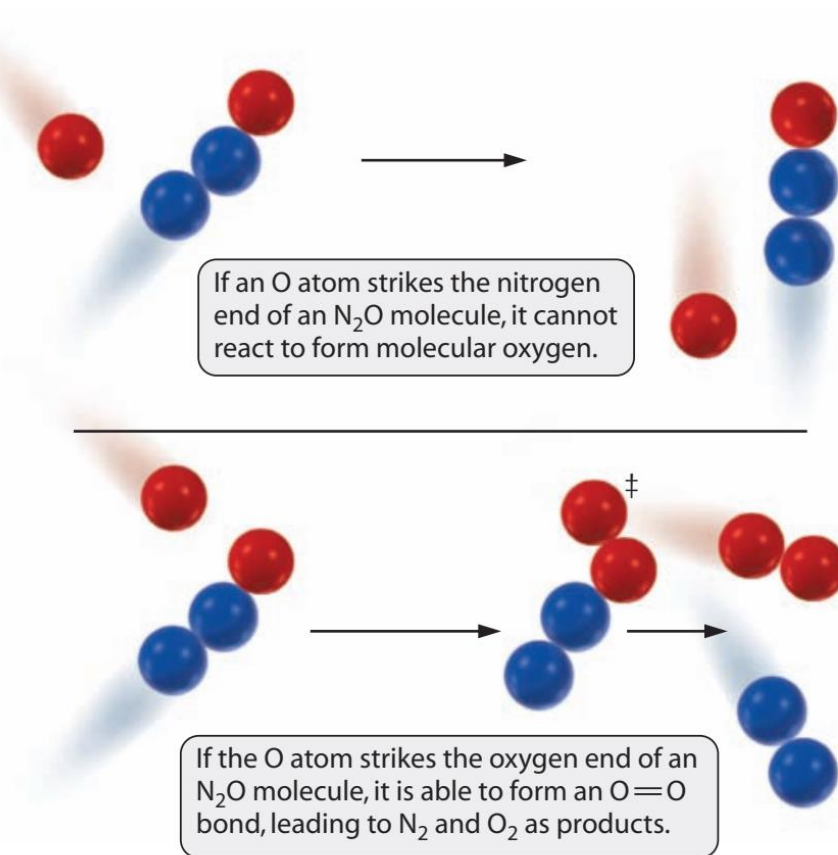


Figure 11.10 ■ The two curves show the Maxwell-Boltzmann distribution of molecular speeds for two temperatures. The shaded areas represent the number of molecules traveling fast enough to collide with sufficient energy to react. Note that there are many more fast moving molecules at higher temperature.

Role of Geometry (steric factor)

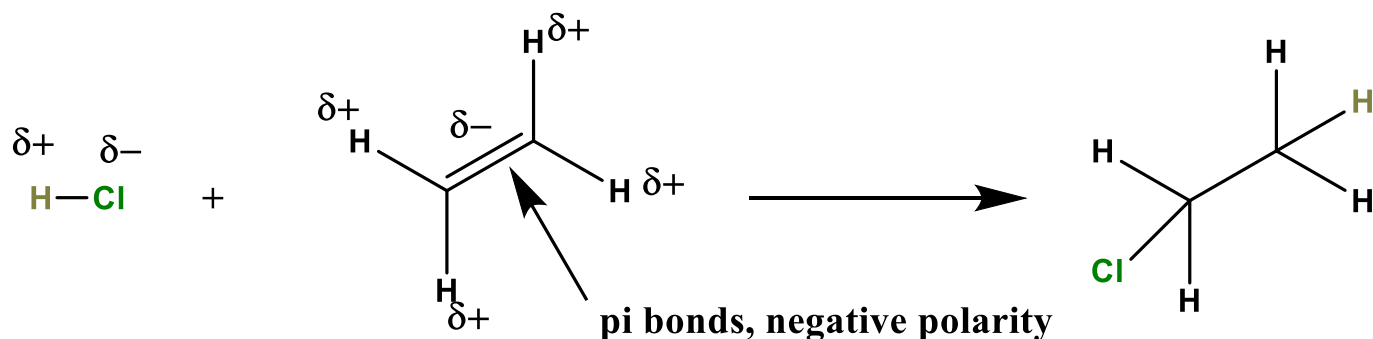


Non-reactive molecular collision
(Unproductive)

Reactive molecular collision
(Productive)

Figure 11.12 ■ Geometric factors can also be important in determining whether a molecular collision is reactive. For the oxygen atom to react with N_2O , it must strike the oxygen end of the molecule. The importance of collision geometry depends on the shapes of the molecules in the reaction.

Markovnikov Addition Reactions

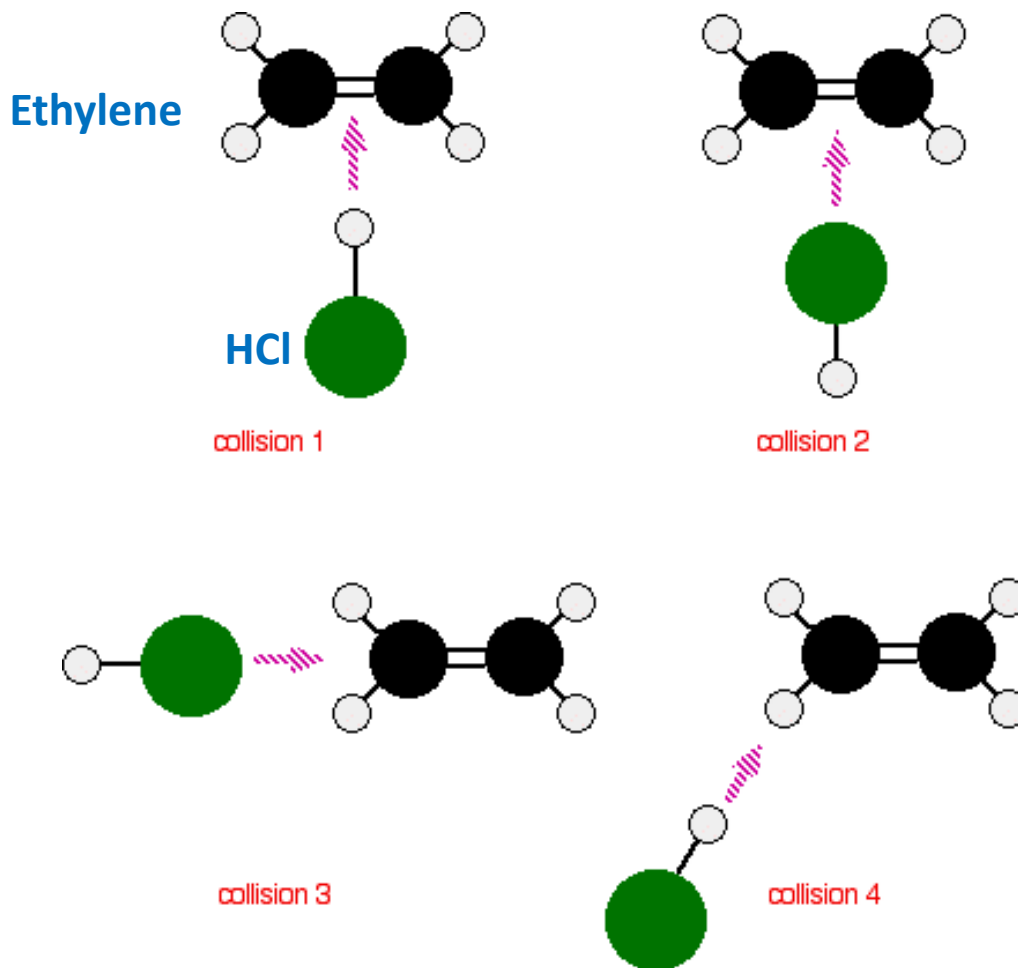


Of 4 possible collision between HCl and Ethene (*an event in Markovnikov addition reactions*), which one may possibly lead on to a reaction?

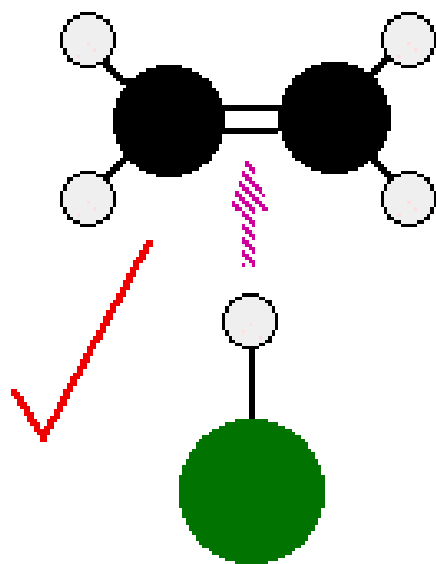
Markovnikov addition reaction

Of 4 possible collision between HCl and Ethene (*an event in Markovnikov addition reactions*), which one may possibly lead on to a reaction?

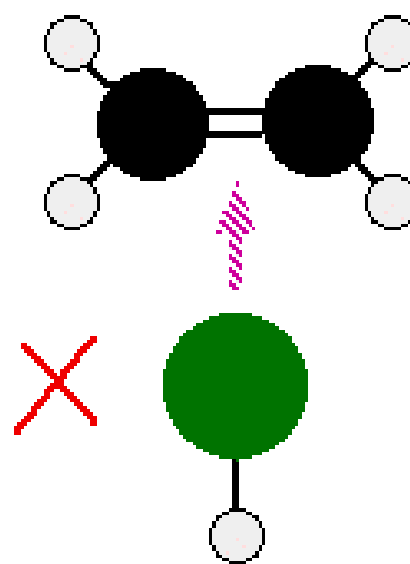
Geometries of Possible collisions between **HCl** and ethylene



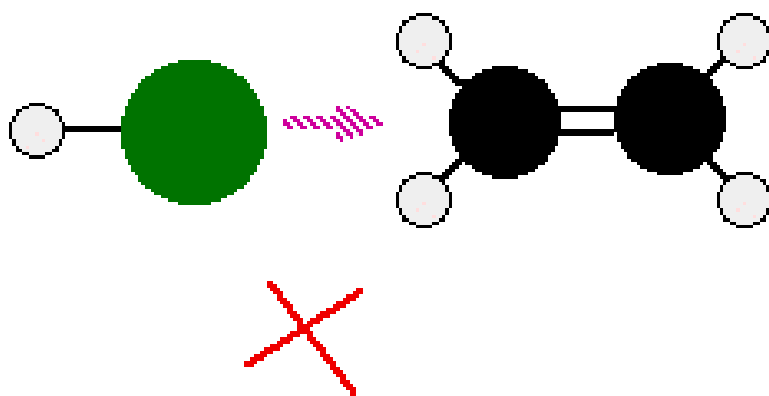
Solution



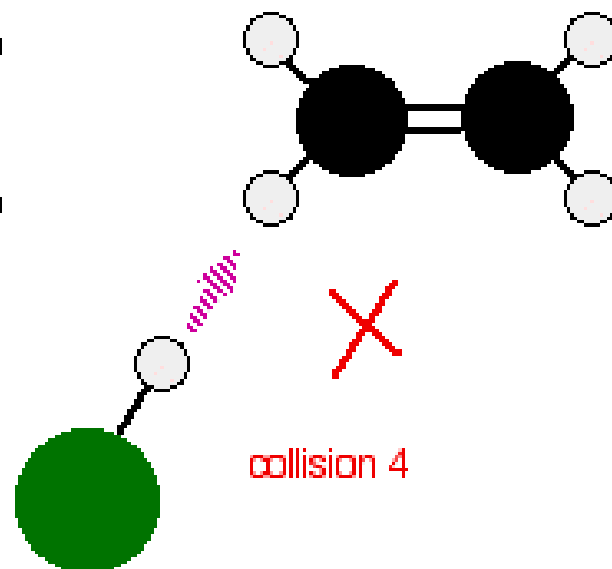
collision 1



collision 2



collision 3



collision 4

Reasoning

Generally, E_a is close to the *average bond energies* of various bond types that are being broken during the reaction.

Average Bond Enthalpies (kJ/mol)

Single Bonds

C—H	413	N—H	391	O—H	463	F—F	155
C—C	348	N—N	163	O—O	146		
C—N	293	N—O	201	O—F	190	Cl—F	253
C—O	358	N—F	272	O—Cl	203	Cl—Cl	242
C—F	485	N—Cl	200	O—I	234		
C—Cl	328	N—Br	243			Br—F	237
C—Br	276			S—H	339	Br—Cl	218
C—I	240	H—H	436	S—F	327	Br—Br	193
C—S	259	H—F	567	S—Cl	253		
		H—Cl	431	S—Br	218	I—Cl	208
Si—H	323	H—Br	366	S—S	266	I—Br	175
Si—Si	226	H—I	299			I—I	151
Si—C	301						
Si—O	368						

Multiple Bonds

C=C	614	N=N	418	O ₂	495
C≡C	839	N≡N	941		
C=N	615			S=O	523
C≡N	891			S=S	418
C=O	799				
C≡O	1072				

Catalysis

Catalysis

Catalysis is the process of increasing the rate of a chemical reaction by adding a substance known as a catalyst.

❖ **Un-catalyzed reaction** $A + B \longrightarrow \text{product}$ [slower]

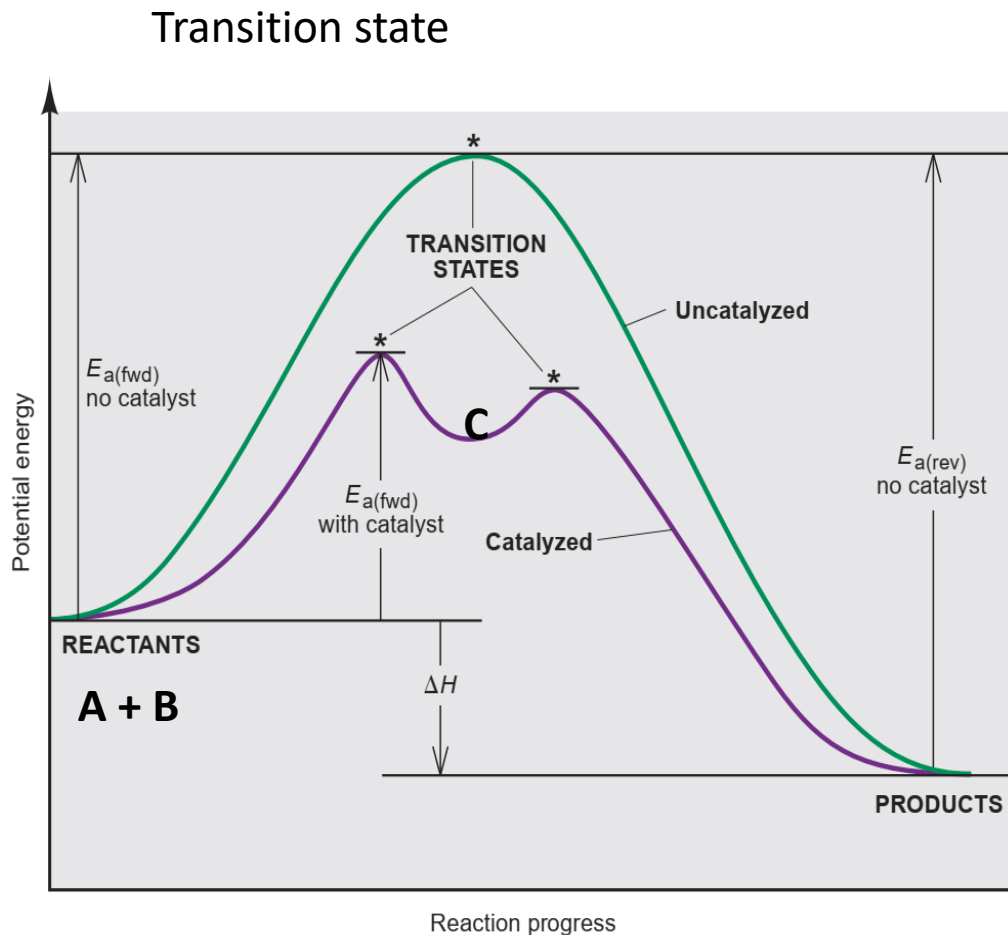
❖ **Catalyzed reaction**



During catalysis ***a reactant-catalyst complex is formed***, for example C in the above reaction

How a catalyst increases the rate of a reaction?

Reaction Progress Diagram (Reaction coordinate)

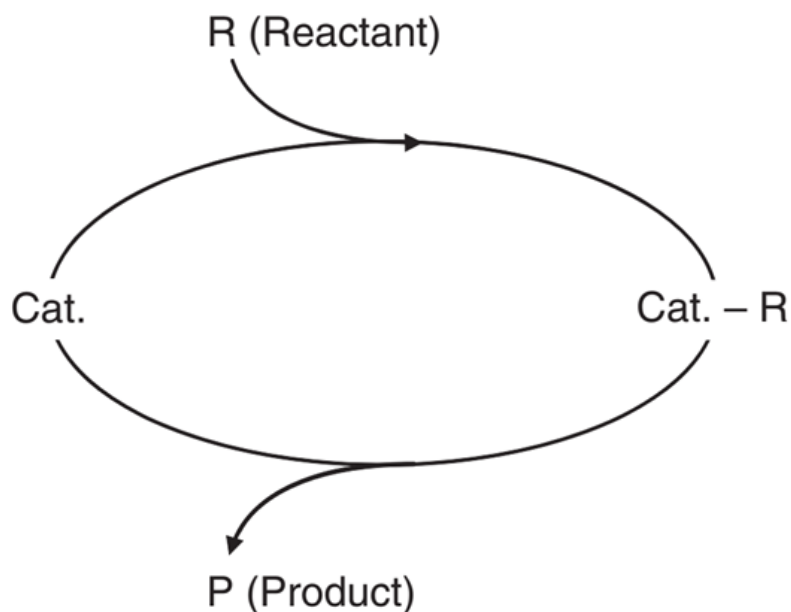


A catalyst provides an **alternative reaction path** with lower activation energy barrier (E_a)

A catalyst only **lowers the E_a** , without altering the **thermodynamic parameters** (ΔH , ΔG , ΔS , K , and **product yield**).

Catalytic Cycle

It is representation of all elementary steps involved in a catalytic reaction in a cyclic loop.



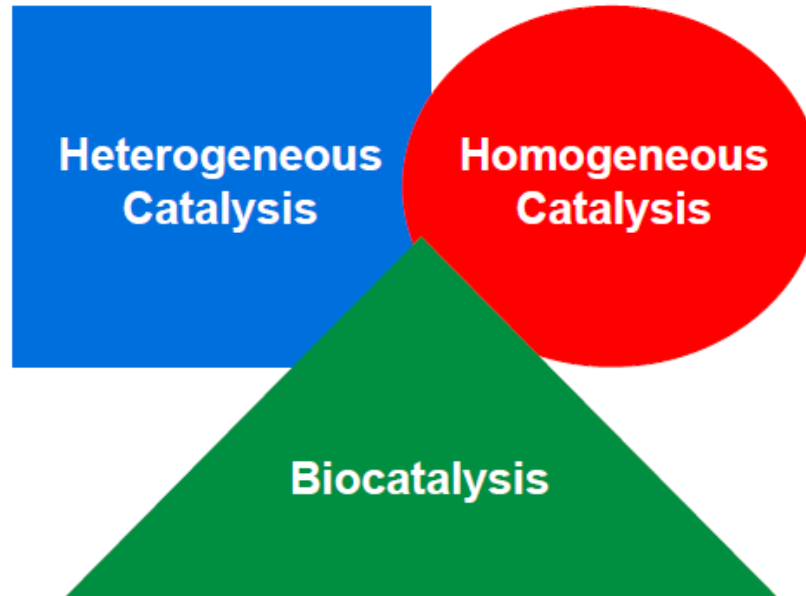
A catalyst is not consumed during the catalysis,
So it will appear only in elementary steps.
It will not appear in reactant or product side of
sum of the elementary reactions

So, one can write the catalyst name above the
net reaction arrow.



Different Types of Catalysis

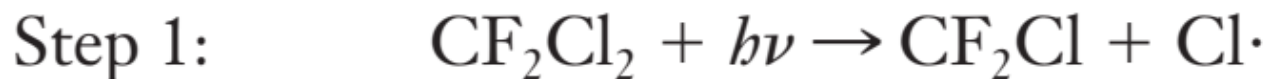
Reactants, products, and catalyst are present in *different phase*.



Reactants, products, and catalyst are present in *same phase*.

Generally homogenous catalysis, catalyzed by *enzymes*

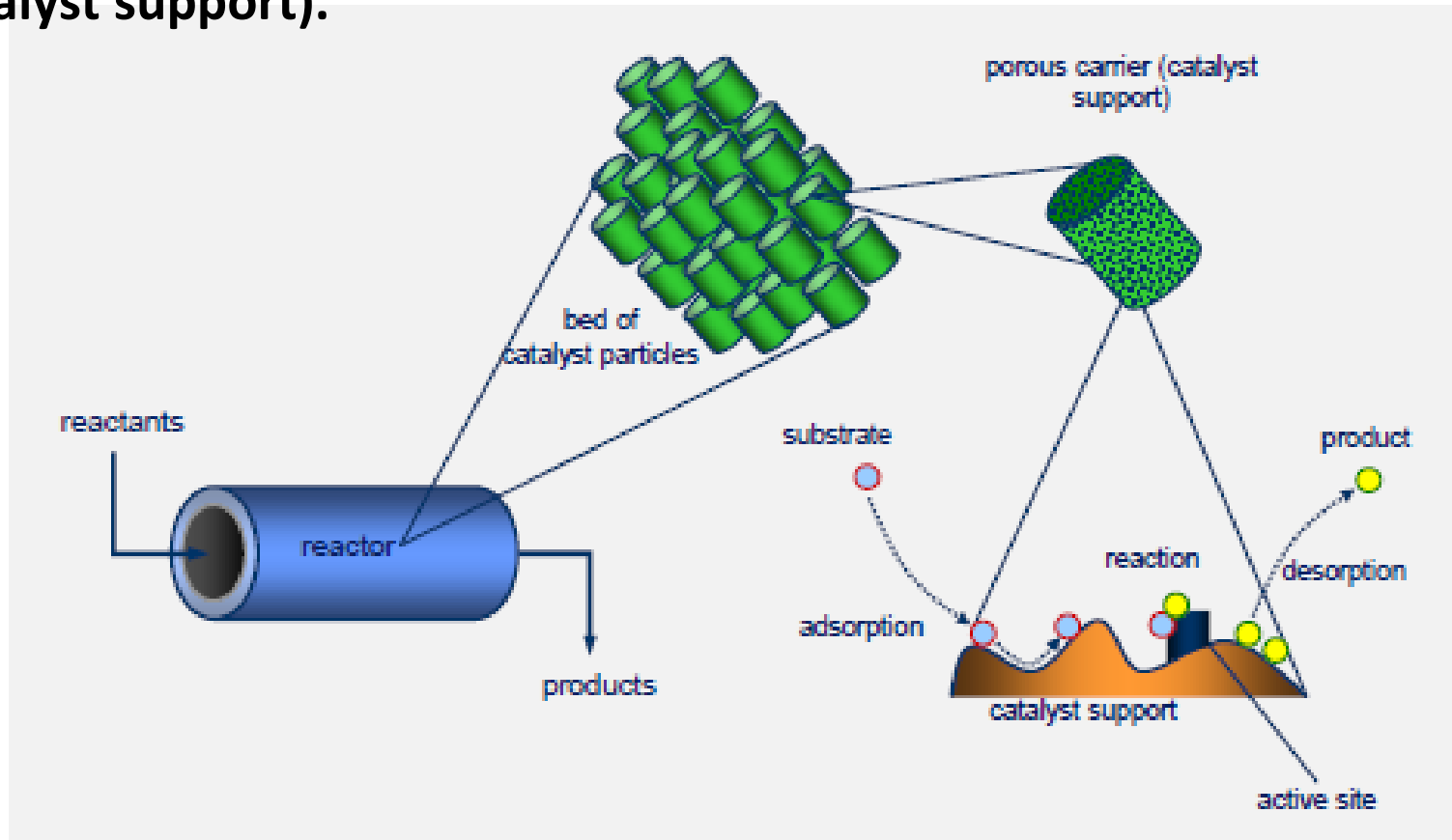
Example of a Gas Phase Homogeneous Catalysis



The catalytic destruction of ozone in the stratosphere in presence of small amount of catalytic species $\text{Cl}\cdot$, chlorine radical formed from CFCs.

Heterogeneous Catalytic Process-Mechanism

Generally reaction occurs on a **solid surface**, where the catalyst is loaded on a thermally stable ceramic material known as monolith (catalyst support).

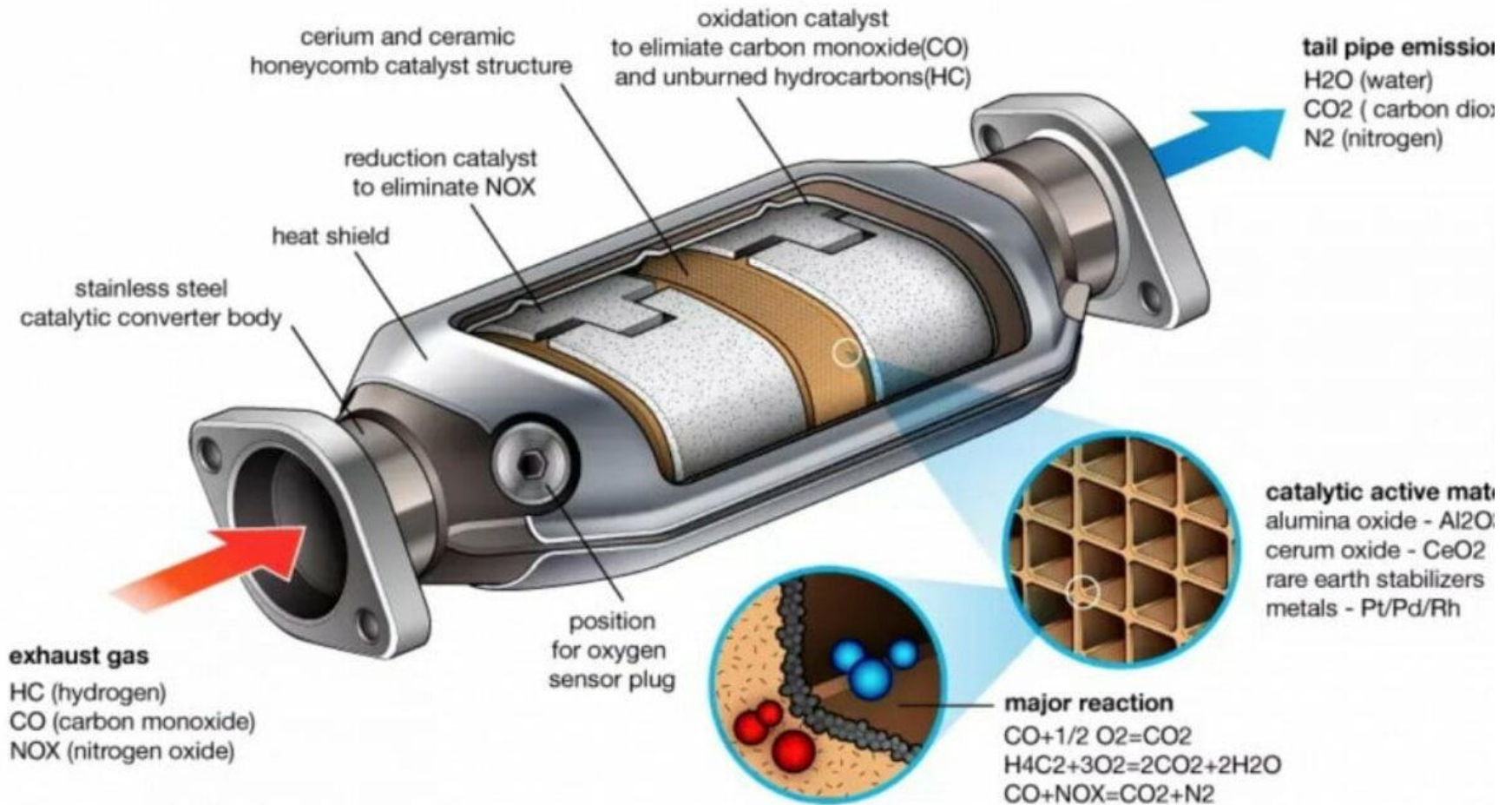


Heterogeneous Catalytic Process-Mechanism

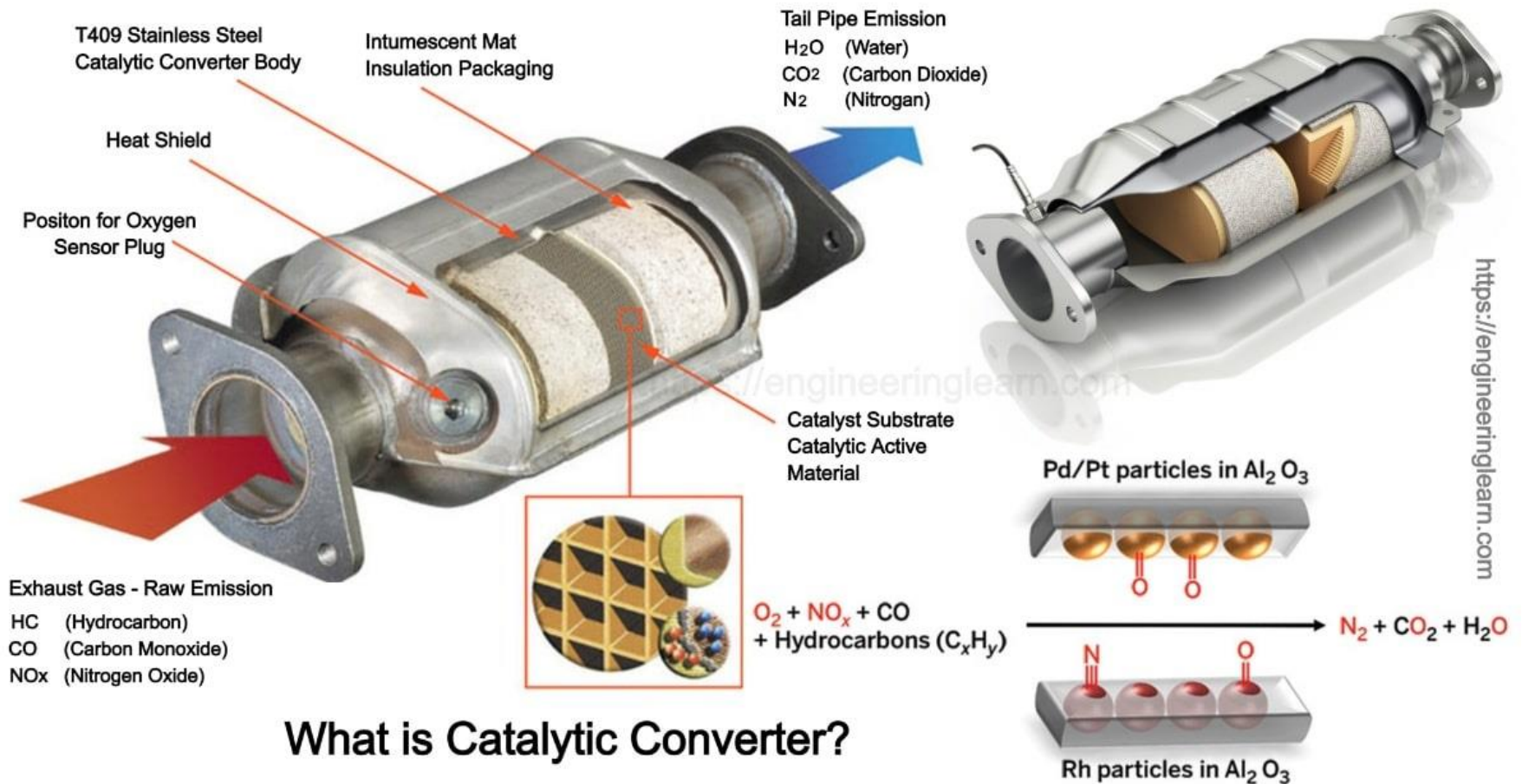
Mechanistic steps

1. **Collision** of a reactant on the catalyst surface
2. **Adsorption** of a reactant on the catalyst surface
3. **Dissociation** of reactant on catalyst surface
4. Product formation
5. **Desorption** of the product from the catalyst surface

Catalytic Converters



Platinum and Rhodium



What is Catalytic Converter?

Decomposition of NO on Pt Surface ($2\text{NO} \rightarrow \text{O}_2 + \text{N}_2$)

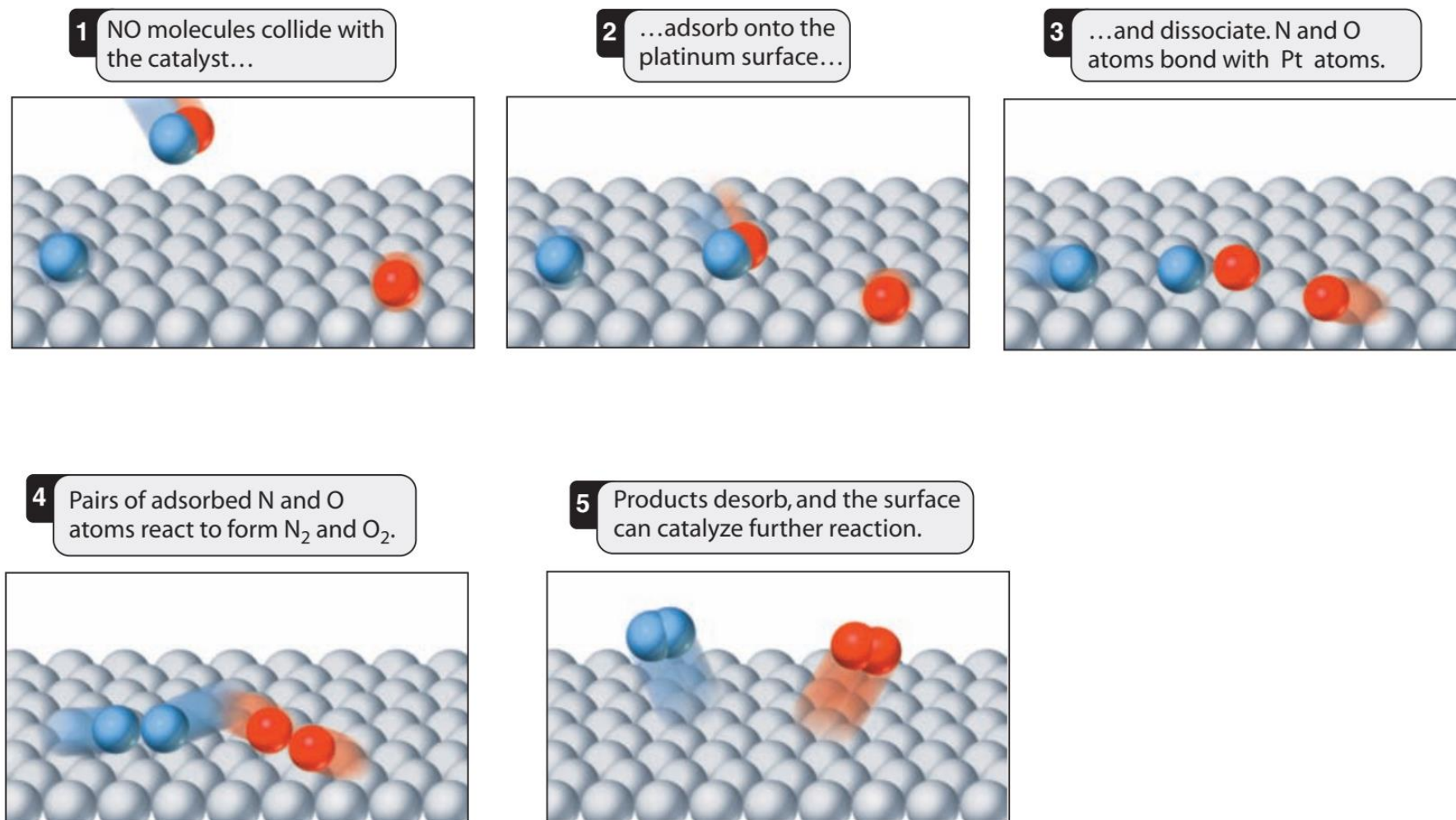
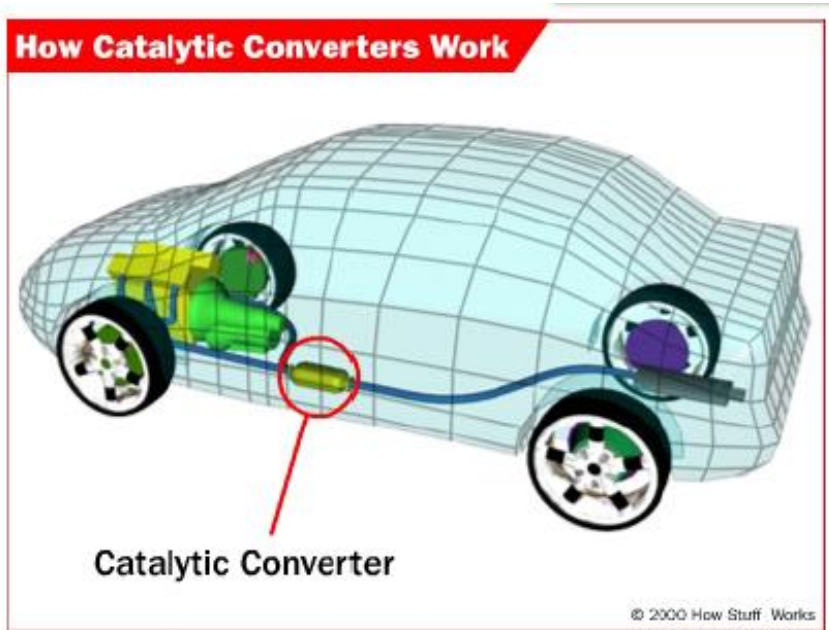


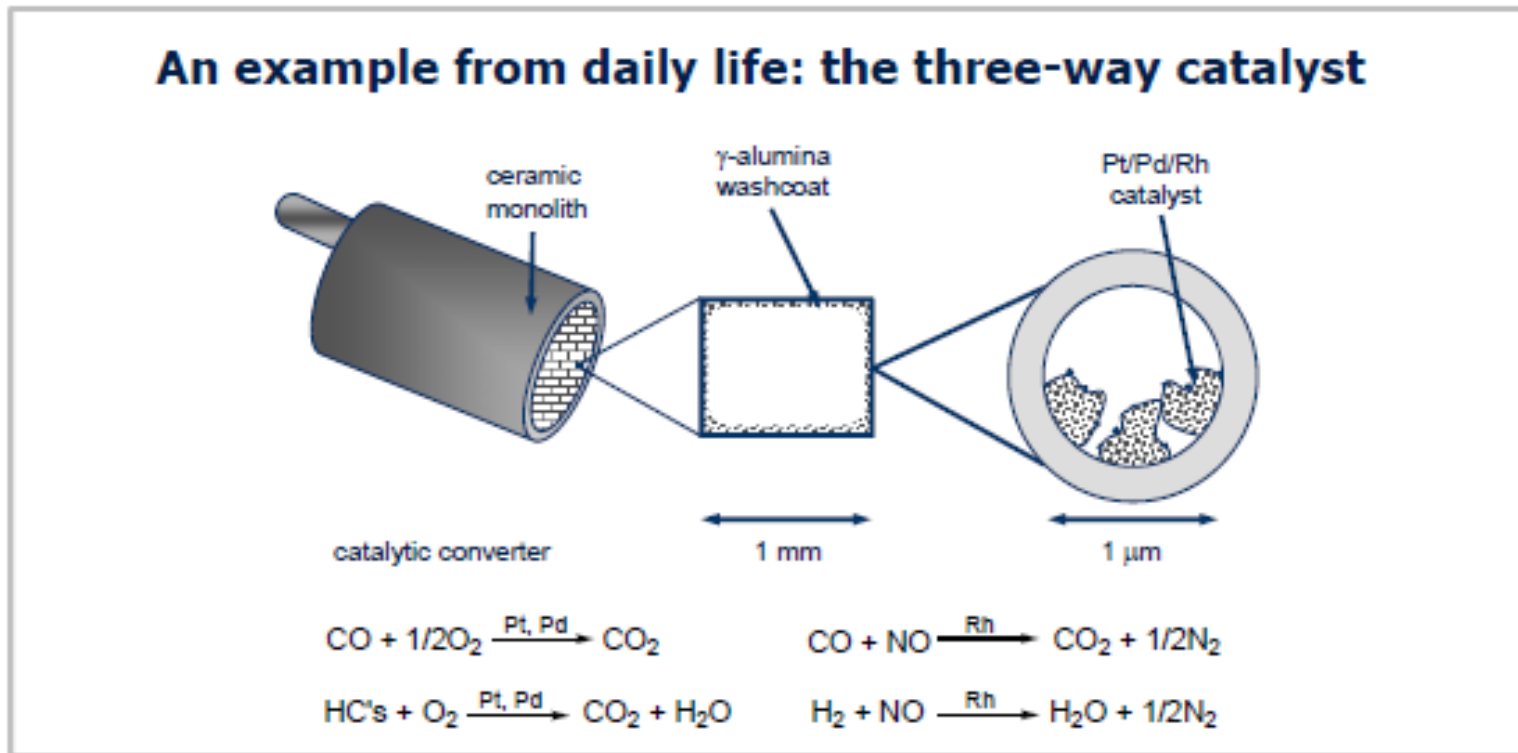
Figure 11.15 ■ A schematic model of the steps involved in the catalytic removal of NO from exhaust gas.

Catalytic converter in a car exhaust



Catalytic converters in automobiles are filled with a porous ceramic material that contains three metals, Pd, Pt, and Rh. These metals catalyze the removal of carbon monoxide (CO) and nitrogen oxides (NO_x) and residual hydrocarbons (HC's) from the exhaust. Since they take care of three obnoxious gases, the catalyst is called a three-way automotive catalyst.

Catalytic converter in a car exhaust



Catalytic converters in automobiles are filled with a porous ceramic material that contains three metals, **Pd, Pt, and Rh**. These metals catalyze the **removal of** carbon monoxide (**CO**) and nitrogen oxides (**NO_x**) and residual hydrocarbons (**HC's**) from the exhaust. Since they take care of three obnoxious gases, the catalyst is called a three-way automotive catalyst.

Haber Bosch Process - Ammonia Synthesis (NH_3)

Ammonia (NH_3) is an exceedingly useful and *important chemical on an industrial basis*.

Annual global production exceeds **120 million tons**.

Nitrogen fertilizers are produced from ammonia as well as a host of other nitrogen containing chemicals.

The production of ammonia *consumes more than $3 \times 10^{17} \text{ J}$* which represents more than **1 %** of global energy consumption.

Therefore, the *discovery of a more efficient and lower temperature ammonia* synthesis would have profound implications not only for fossil fuel consumption, but also the world-wide economy.

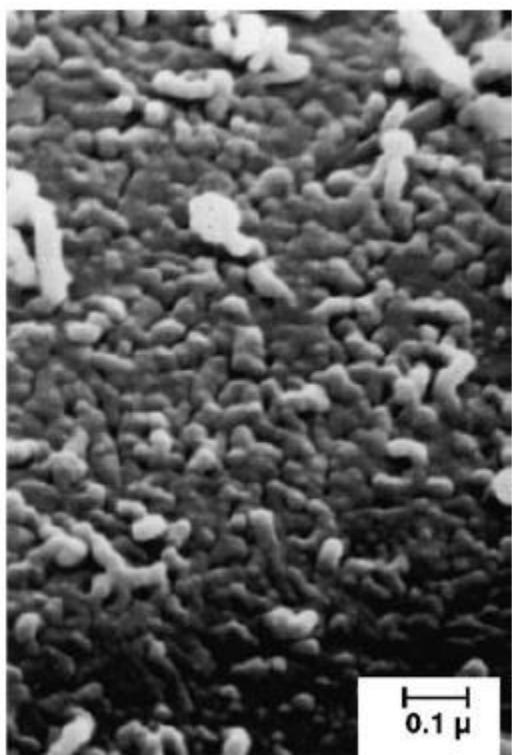
Haber Bosch Process - Ammonia Synthesis (NH₃)



The Haber-Bosch process operates at **200–300 bar** and **670–770 K** over an **Fe/K/CaO/ Al₂O₃** catalyst.

Al₂O₃ is the preferred substrate additive because it acts as a **structural promoter**, ensuring high dispersion of the **catalytically active Fe clusters** and hindering their tendency to sinter into larger particles. The **CaO** may assist in this process.

The **K** acts as an **electronic promoter**, modify the band gap of the catalyst to provide the desirable effect.



Transmission electron microscopic
image of the catalyst showing porous structure with iron clusters

Haber Bosch Process - Ammonia Synthesis (NH₃)

Elementary steps in ammonia synthesis



* : Refers surface **active site** of the catalyst



Step 1 – **adsorption** of N₂ on active site

Step 2 – **Dissociation** of N₂ on surface of the catalyst

Step 3 – Combining of H and N on the surface of catalyst

Step 4 - Combining of NH and H

Step 5 – Combining of NH₂ with H

Step 6 – Release of ammonia from surface active site

Step 7 – Shows the dissociation of H₂ on the surface

Iron is the very effective catalyst for breaking the strong bond between nitrogen atoms with a bond enthalpy of **941 kJ/mol**.

It is the **rate** limiting or **determining step**.

This process lowers the activation energy barrier for the reaction.



Fischer-Tropsch Synthesis

(Production of Synthetic Fuels)

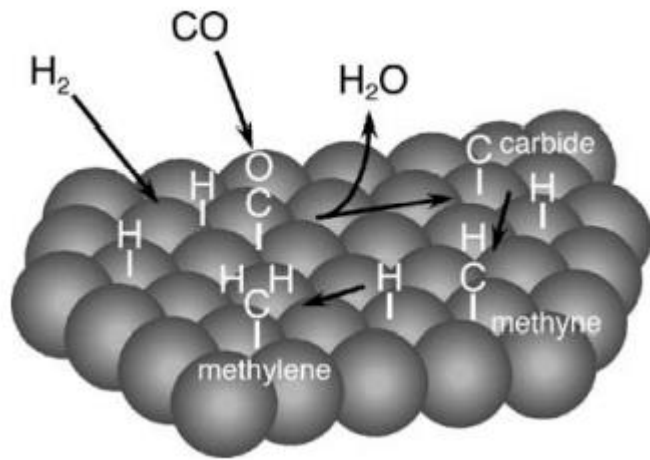
Fischer-Tropsch chemistry proceeds via a complex set of reactions, which consume **CO** and **H₂** and produce a mixture of alkanes **C_nH_{2n+2}**, alkenes C_nH_{2n}, alcohols C_nH_{2n+1}OH and other oxygenated compounds, aromatics as well as CO₂ and H₂O.

Some examples of these reactions are



Fischer-Tropsch Synthesis (Production of Synthetic Fuels)

Catalyst Composition-Product Formation



The use of **CuO/ZnO/Al₂O₃** or **CuO/ZnO** catalyst allows the process to be run at the relatively mild temperature of **230–270 °C**.

When **Cu** is the active site, the selectivity of the catalyst is remarkably high for the production of methanol.

Although higher alcohols, ethers and alkanes are thermodynamically preferred, the selectivity for methanol is often **> 99 %**.

In these catalysts the Cu acts as the active site. The ZnO provides a matrix into which it can dissolve and the Al₂O₃ ensures a high surface area (dispersion) for the catalyst.

Product selectivity depends on metal

A **Co** or **Fe** catalyst is best for producing the **low molecular weight** hydrocarbons preferred as liquid fuels (both petrol and diesel)

whereas, a **Ru** catalyst is selective for producing **high molecular weight** waxy hydrocarbons.

Rh-based catalysts give increased yields of oxygenates, particularly **methanol and ethanol**.

A **Ni** catalyst is the best for formation of **methane**.

Table 16.7 Some Modern Processes That Employ Catalysts

Reactants	Catalyst	Product	Use
Homogeneous			
Propylene, oxidizer	Mo(VI) complexes	Propylene oxide	Polyurethane foams; polyesters
Methanol, CO	$[\text{Rh}(\text{CO})_2\text{I}_2]^-$	Acetic acid	Poly(vinyl acetate) coatings; poly(vinyl alcohol)
Butadiene, HCN	Ni/P compounds	Adiponitrile	Nylons (fibers, plastics)
α -Olefins, CO, H ₂	Rh/P compounds	Aldehydes	Plasticizers, lubricants
Heterogeneous			
Ethylene, O ₂	Silver, cesium chloride on alumina	Ethylene oxide	Polyesters, ethylene glycol, lubricants
Propylene, NH ₃ , O ₂	Bismuth molybdates	Acrylonitrile	Plastics, fibers, resins
Ethylene	Organochromium and titanium halides on silica	High-density polyethylene	Molded products

Parameters to Compare Different Catalysts

Turnover Number (TON) is defined as the number of reaction cycle that can occur per catalyst binding site (**at any time interval**).

Turnover frequency (TOF) is defined as the number of catalytic cycles occurring per unit time.

$$\text{TOF} = \frac{\text{volumetric rate of reaction}}{\text{number of centers/volume}} = \frac{\text{moles}}{\text{volume} \cdot \text{time}} \frac{\text{volume}}{\text{moles}} = \text{time}^{-1}$$

Example for TON and TOF

Turn over number (TON) - Number of reactions a **single site** can achieve

e.g. 1 mmol Pd converts 1000 mmols of $\text{CO} \rightarrow \text{CO}_2$

$$\text{TON} = 1000$$

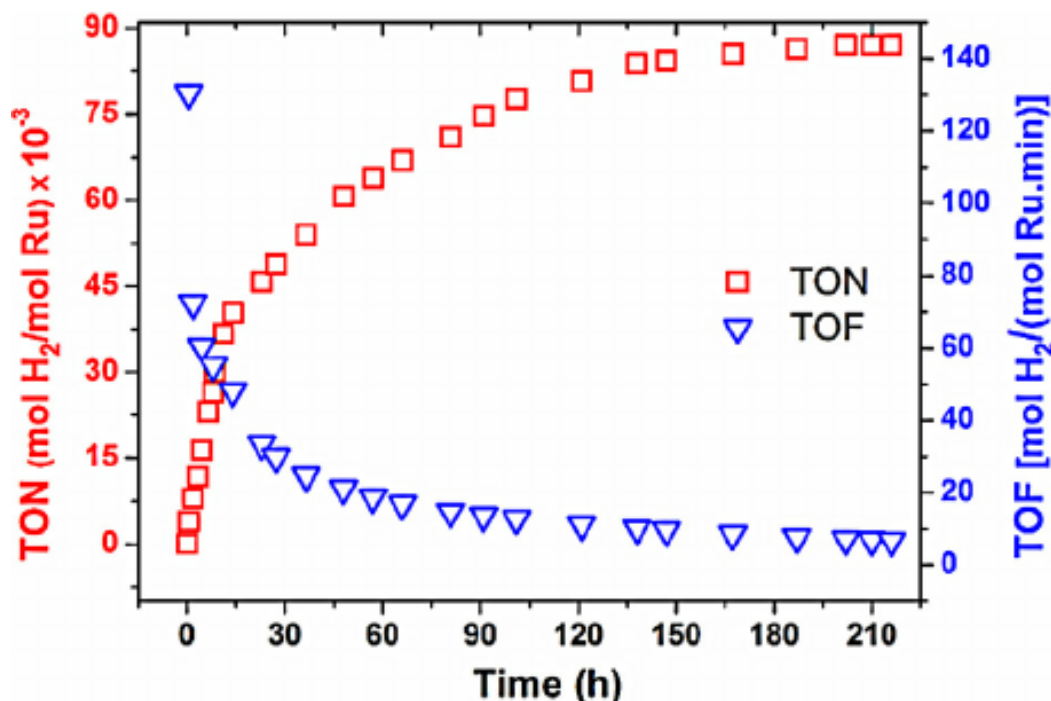
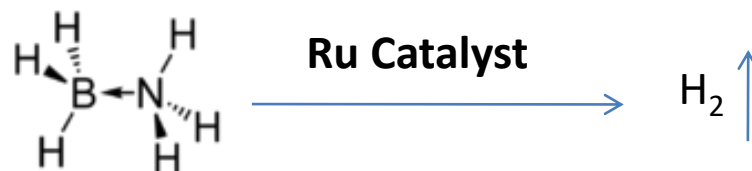
Turn over frequency (TOF) - Number of reactions **per site per unit time**.

e.g. 1 mmol Pd converts 1000 mmols of $\text{CO} \rightarrow \text{CO}_2$ in 10 s

$$\text{TOF} = 100 \text{ s}^{-1}$$



Hydrogen Evolution From Ammonia-Borane



Selectivity (in %)

The ability of the **catalyst** to direct a reaction to yield a particular product is referred to as the **selectivity** of the **catalyst**.



$$\text{Selectivity} = \frac{\text{Number of moles of desired product}}{\text{Total number of moles of all products}} \times 100$$

Specificity

Specificity of a catalyst refers to the specific reaction it can catalyze (e.g. Reactant specificity)

Higher the selectivity and specificity better the catalyst

Enzyme Catalysis

- **Enzymes** are macromolecular **biological catalysts** that accelerate chemical reactions

PROTEIN STRUCTURE

Scaffold to support and position active site

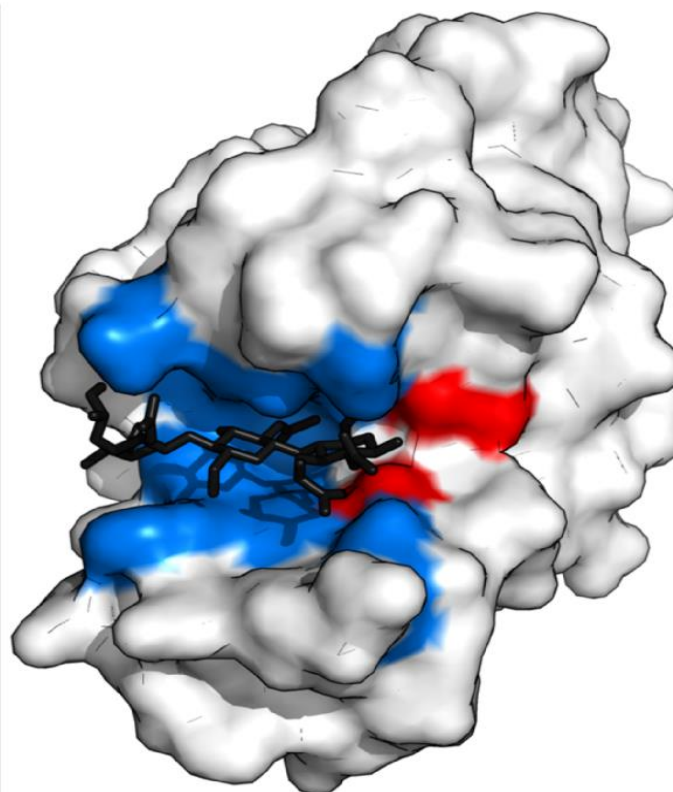
ACTIVE SITE

BINDING SITES

Bind and orient substrate(s)

CATALYTIC SITE

Reduce chemical activation energy



Enzymes

Enzymes are the catalysts of nature

With the exception of catalytic RNA, **all enzymes are proteins.**

Catalyst alter the rate of a chemical reaction without undergoing a permanent change in structure.

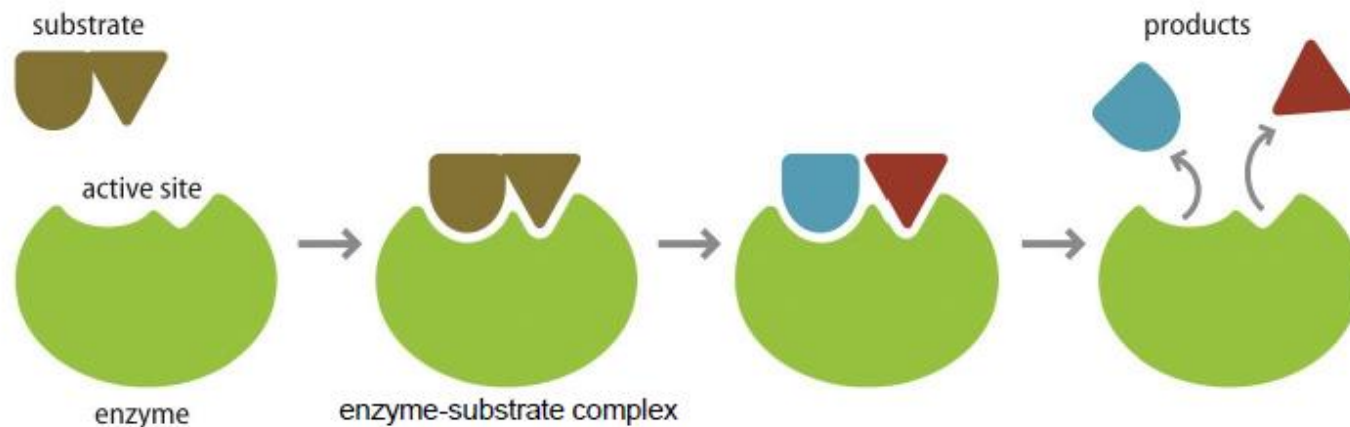
- For example, **curd formation** (rennet coagulation) is a reaction that occurs upon adding the enzyme rennin to milk.
- In this reaction, the **substrate** is a **milk protein** (e.g., casein) and the **enzyme** is **rennin**.
- The **products** are two **polypeptides** that have been formed by the cleavage of the larger peptide substrate.

The site of catalytic activity on the enzyme is known as the *active site*.

The molecule that binds to the active site and is acted upon by the enzyme is called the *substrate*.

The two together form what is known as the *enzyme-substrate complex*.

The function of an enzyme is to increase the rate of a chemical reaction without affecting its equilibrium.



Enzyme Catalysis

Lock-and-key model

An enzyme binds a substrate in a region called the **active site**

The active site shape is complementary to the substrate
i.e. not all substrates can fit the active site.

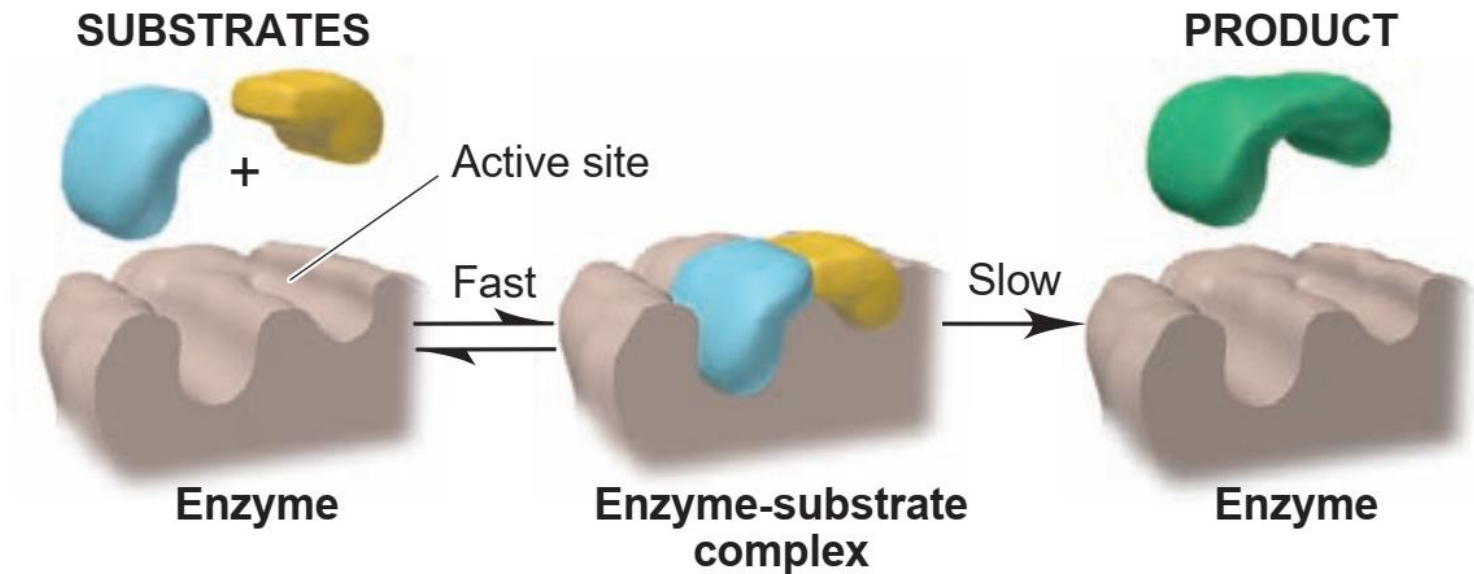
Amino acid sidechains in the active site bind the substrate.

Induced-fit model

Enzyme structure is **flexible**, not rigid.

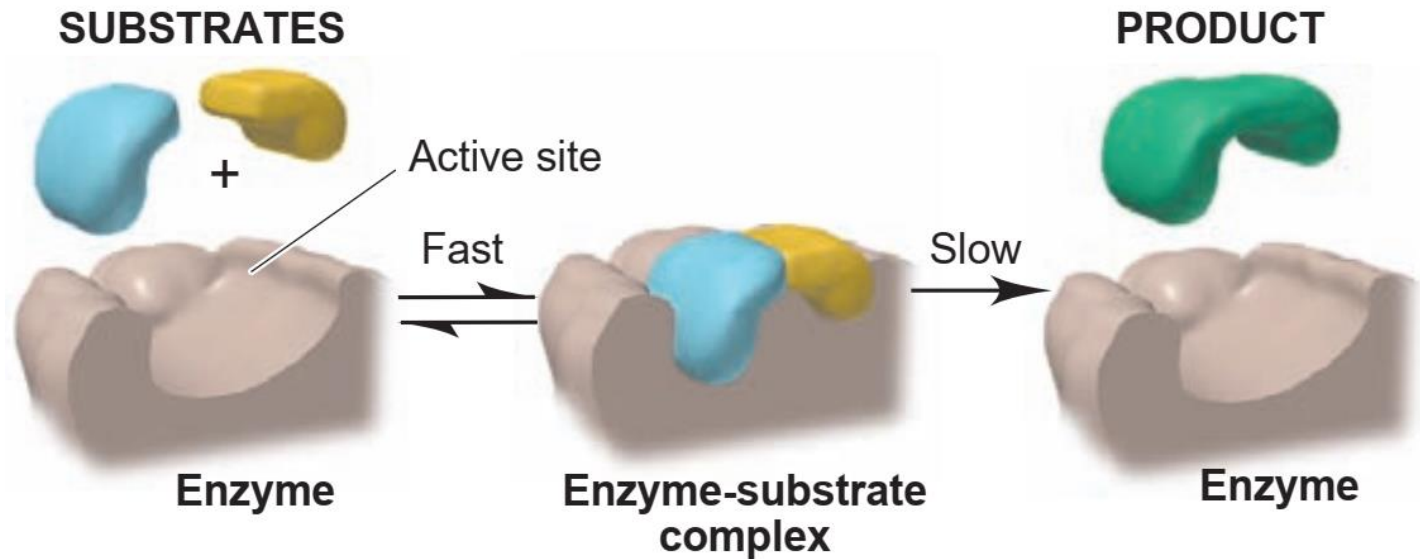
Enzyme and active site adjust their shape to bind the substrate

Lock-and-key model (Emil Fischer 1894)

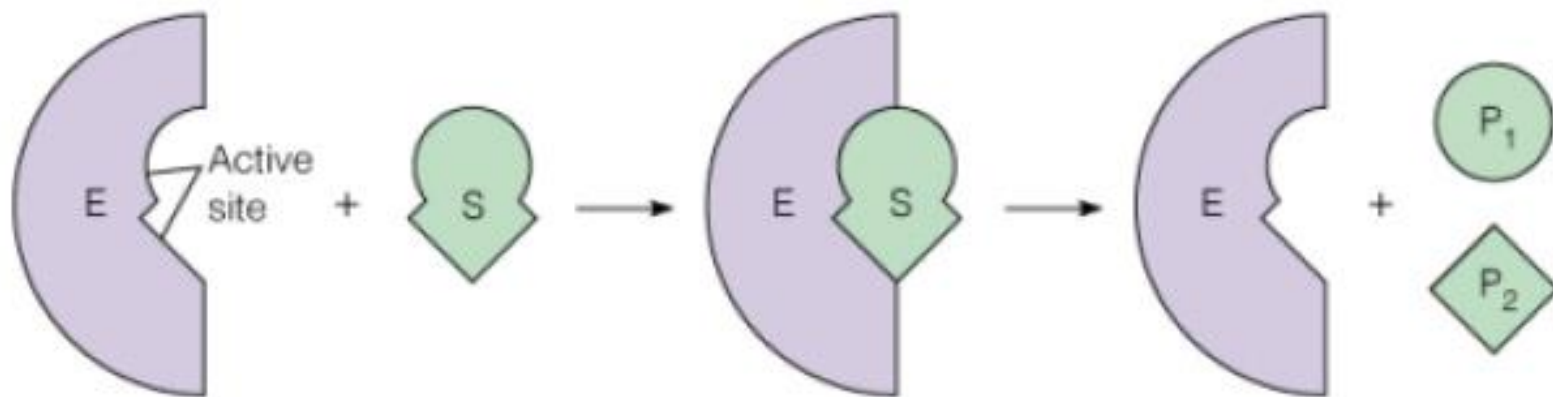


In the lock and-key model, the active site is thought to be an **exact fit** for the substrate shapes.

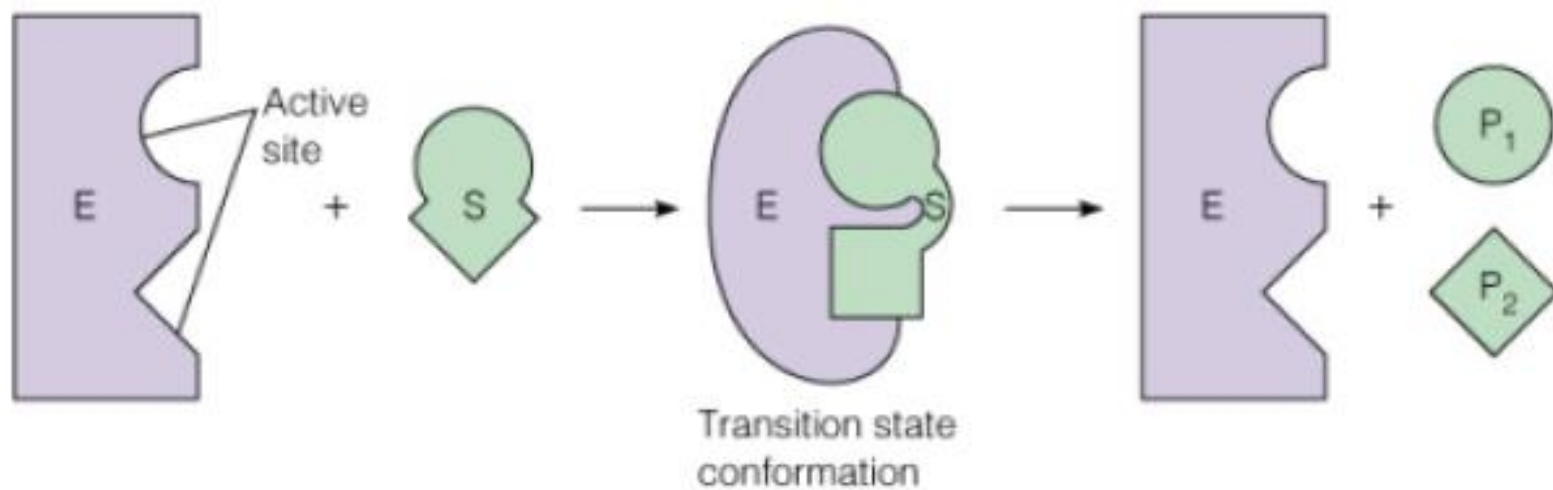
Induced-fit model (Daniel Koshland, 1958)



In the induced-fit model, the active site is thought to change shape to fit the substrates



(a) Lock-and-key model

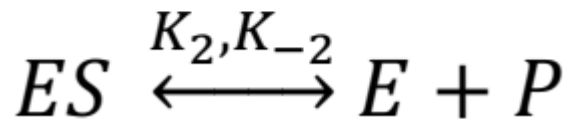
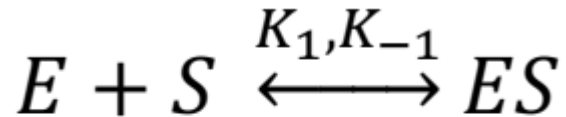


(b) Induced fit model

Enzyme Catalysis

Michaelis - Menten Mechanism

One of the best-known models of enzyme kinetics, named after German biochemist **Leonor Michaelis**, and Canadian physician **Maud Menten** (in the year **1913**)

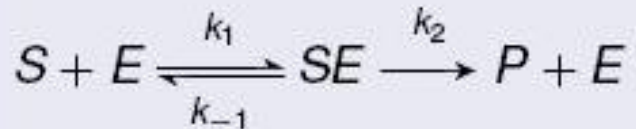


Consider a simple single substrate reaction, where the free **enzyme E** binds to the **substrate S** to form **an enzyme-substrate complex SE (or ES)**, that transforms **S to product P** and dissociates.

Enzyme Catalysis

Michaelis - Menten Mechanism

Quasi-steady state



$$\frac{d[SE]}{dt} \approx 0$$

$$K_m = \frac{k_{-1} + k_2}{k_1} = \frac{[S][E]}{[SE]}$$

$$\frac{d[P]}{dt} \approx \frac{k_2[E]_0 [S]}{[S] + K_m}$$

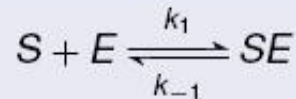
k_1 = Rate constant for the formation of ES

k_{-1} = Rate constant for the dissociation of ES

k_2 = Rate constant for the formation product from ES

K_m = Michaelis Constant, or Dissociation equilibrium constant of ES

Equilibrium



$$\frac{d[SE]}{dt} = 0$$

$$K_m = \frac{k_{-1}}{k_1} = \frac{[S][E]}{[SE]}$$

Enzyme Catalysis

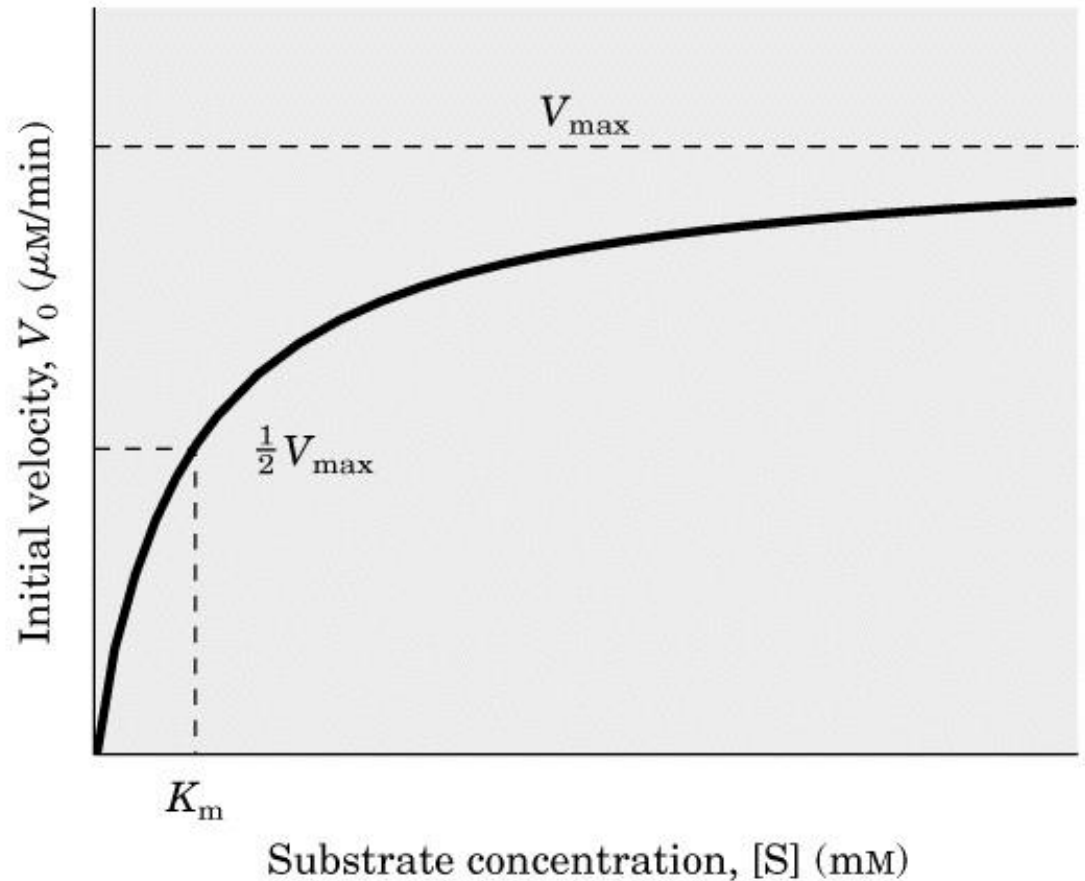
Michaelis - Menten Mechanism

$$\frac{d[P]}{dt} \approx \frac{k_2[E]_0 [S]}{[S] + K_m}$$

$$V_0 = \frac{V_{max}[S]}{K_m + [S]}$$

Michaelis-Menten rate constant, K_m , is defined as

$$\frac{K_{-1} + K_2}{K_1} = K_m$$



Enzyme Catalysis

Michaelis - Menten Mechanism

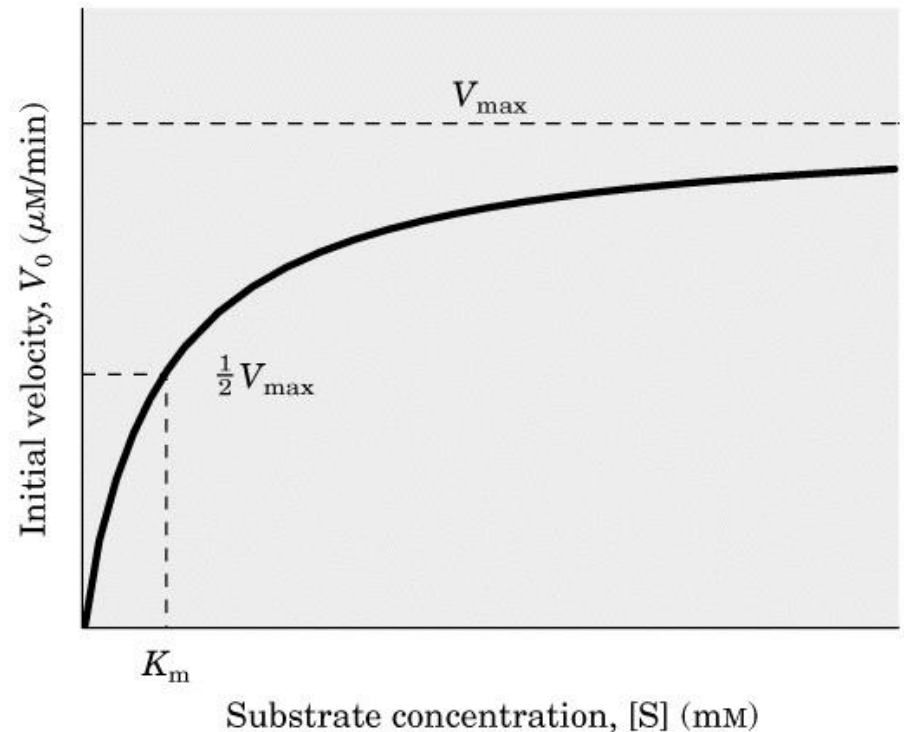
$$V_0 = \frac{V_{max}[S]}{K_m + [S]}$$

$$V_0 = \frac{V_{max}}{2}$$

$$\frac{V_{max}}{2} = \frac{V_{max}[S]}{K_m + [S]}$$

$$\frac{1}{2} = \frac{[S]}{K_m + [S]}$$

$$K_m = [S]$$



Enzyme Catalysis

Michaelis - Menten Mechanism

$$V_o = \frac{V_{max}[S]}{K_m + [S]}$$

Case I When $[S] \ll K_m$

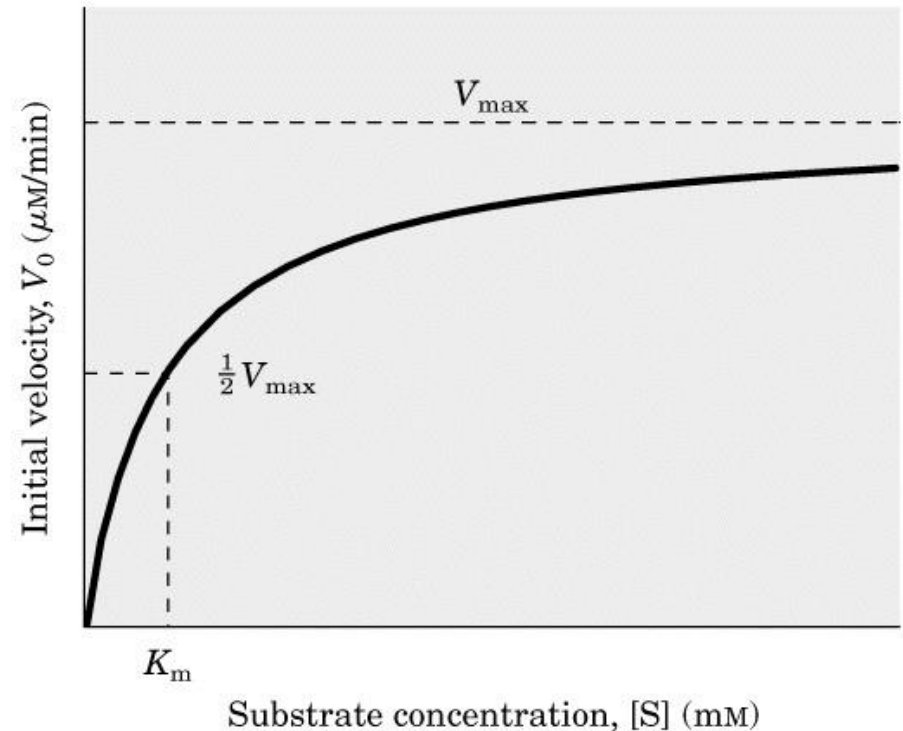
$$V_o = V_{max}[S]/K_m$$

Case II When $[S] = K_m$

$$V_o = \frac{1}{2} V_{max}$$

Case III When $[S] \gg K_m$

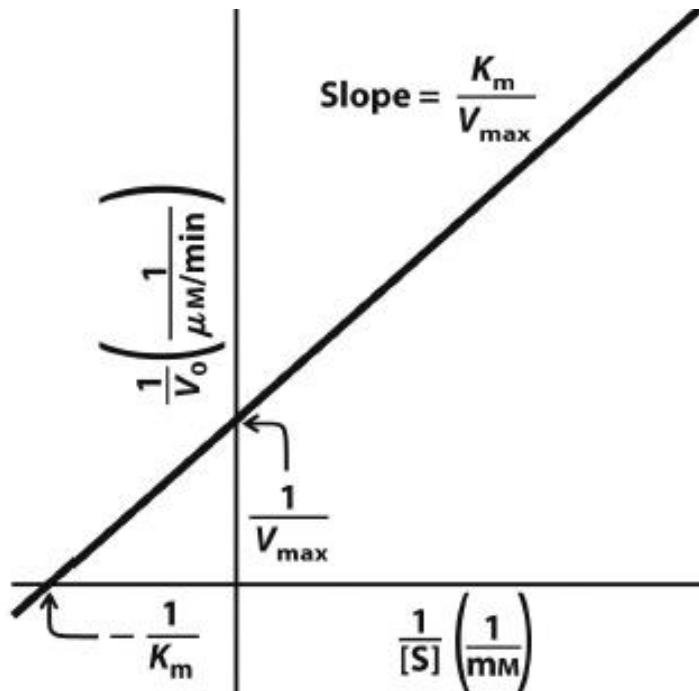
$$V_o = V_{max}$$



Enzyme Catalysis

Michaelis - Menten Mechanism

Lineweaver-Burk plot



$$V_0 = \frac{V_{\max} [S]}{K_m + [S]}$$

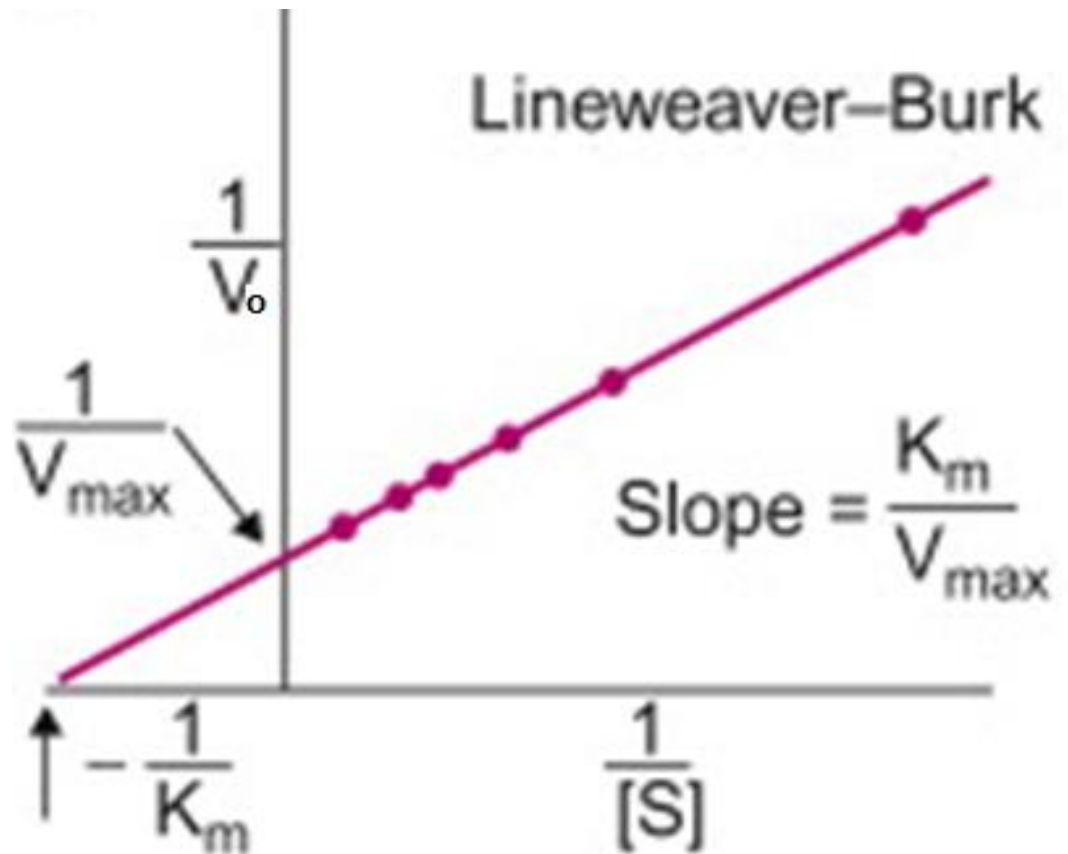
$$\frac{1}{V_0} = \frac{K_m + [S]}{V_{\max} [S]}$$

$$\frac{1}{V_0} = \frac{k_m}{V_{\max} [S]} + \frac{[S]}{V_{\max} [S]}$$

$$\frac{1}{V_0} = \frac{k_m}{V_{\max} [S]} + \frac{1}{V_{\max}}$$

Lineweaver-Burk Plot or Double Reciprocal Plot

$$\frac{1}{V_0} = \frac{k_m}{V_{max} [S]} + \frac{1}{V_{max}}$$



Enzymes are Targets for Drug Molecules

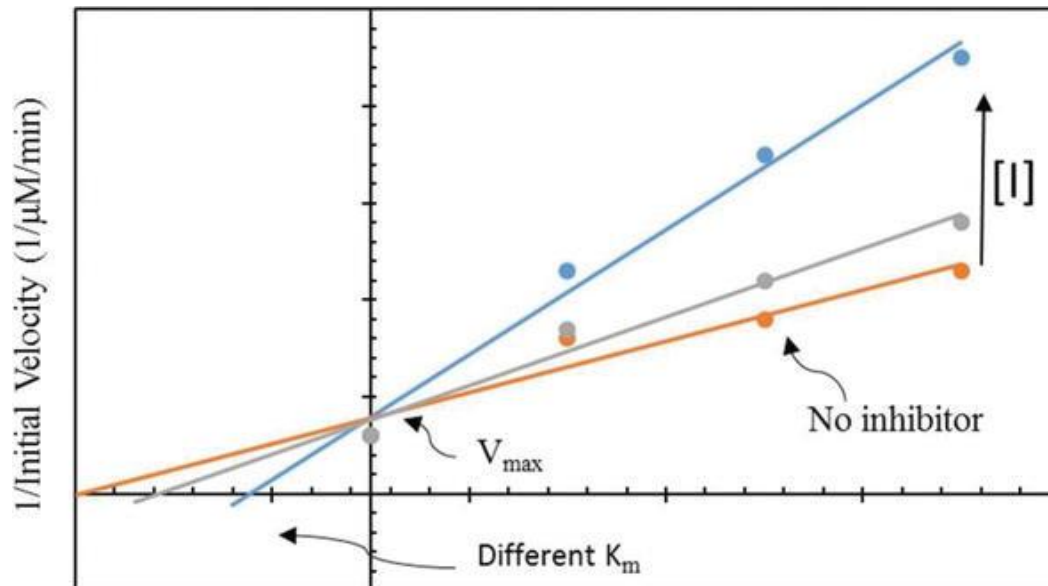
Many drug molecules for the treatment of disease function by inhibiting enzymes.

For example, an important strategy in the treatment of acquired immune deficiency syndrome (AIDS) involves the steady administration of a specially designed *protease inhibitor*.

The drug inhibits an enzyme (**Protease**) that is key to the formation of the protein envelope surrounding the genetic material of the human immune deficiency virus (HIV). Without a properly formed envelope, HIV cannot replicate in the host organism

Competitive Inhibitors

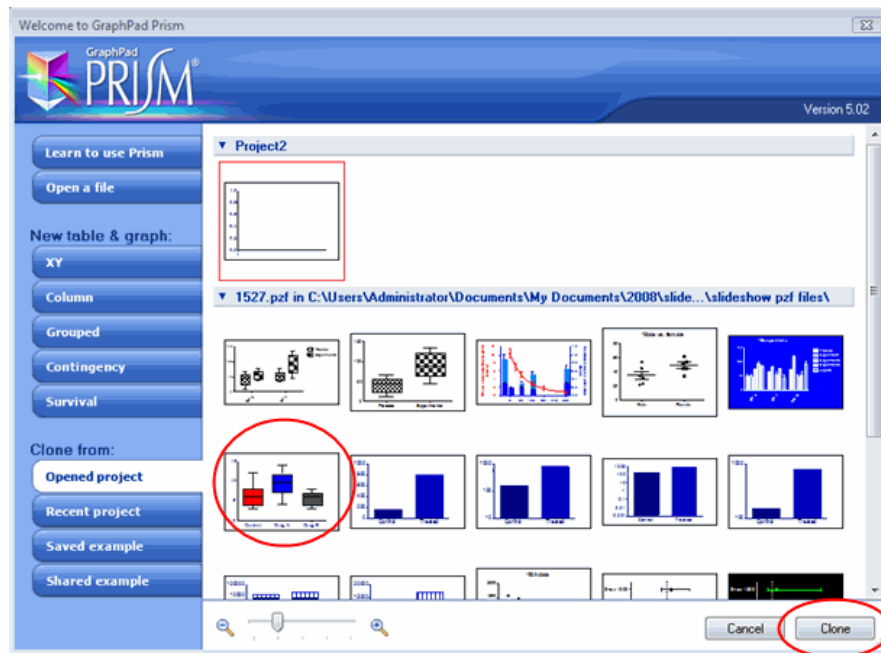
If a drug competes for the enzyme substrate, then drug is known as competitive inhibitor (I).



Slope of the line increases upon adding more and more amount of drug, hence V_{max} decreases.

K_m increases for competitive inhibition

Common Software's used for Enzyme-Kinetic Data Analysis



Problems Based on Kinetics

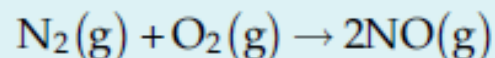
Problem 1

EXAMPLE 9.1: DETERMINING RATES OF REACTION FROM EXPERIMENTAL DATA

Determine the rate of formation of nitric oxide from the reaction of 1 mole of nitrogen gas with oxygen if the nitrogen is found to drop to $\frac{1}{2}$ mole in 0.25 s.

Solution 1

1. Write the balanced chemical equation.



2. Determine the rate of loss of nitrogen gas.

$$-d[\text{N}_2]/dt = -0.5\text{mol}/0.25\text{s} = 2.0 \text{ mol} \bullet \text{s}^{-1}$$

3. Determine the rate of formation of NO.

According to the chemical equation, 2 moles of NO are formed for each mole of N_2 lost.

$$-d[\text{N}_2]/dt = \frac{1}{2}d[\text{NO}]/dt = 2 \times 2\text{mol} \bullet \text{s}^{-1} = 4\text{mol} \bullet \text{s}^{-1}$$

Problem 2

At high temperatures, nitrogen dioxide decomposes to nitric oxide and oxygen.



Experimental data for the reaction at 300°C and four initial concentrations of NO_2 are listed in the following table:

Experiment	$[\text{NO}_2]_0$ (M)	Initial Rate (M/s)
1	0.015	1.22×10^{-4}
2	0.010	5.40×10^{-5}
3	0.0080	3.46×10^{-5}
4	0.0050	1.35×10^{-5}

Determine the reaction order and the rate constant

$$\text{Rate} = k[\text{NO}_2]^2$$

$$1.22 \times 10^{-4} \text{ M/s} = k [0.015]^2$$

$$k = 1.22 \times 10^{-4} / [0.015]^2 \text{ M}^{-1} \text{ s}^{-1}$$

Problem 3

EXAMPLE 9.2: DETERMINING REACTION ORDER FROM EXPERIMENTAL MEASUREMENTS OF REACTION RATE WITH CHANGING REACTANT CONCENTRATIONS

Determine the reaction order for each reactant, the overall reaction order, and the rate law for the reaction of nitrogen oxide with bromine gas from the following initial rate measurements:

Experiment	[NO] (M)	[Br ₂] (M)	Rate of Reaction (M • s ⁻¹)
(1)	0.2	0.2	4.0×10^{-4}
(2)	0.4	0.2	1.6×10^{-3}
(3)	0.2	0.1	2.0×10^{-4}

Solution 3

1. Determine the reaction order for NO.

Comparing experiments 1 and 2: the NO concentration is increased by a factor of 2 and the Br₂ is constant. The reaction rate increases by a factor of 4.

This means that the reaction is second order in NO.

2. Determine the reaction order for Br₂.

Comparing experiments 1 and 3: the concentration of NO is constant and the concentration of Br₂ is decreased by a factor of 2. The reaction rate is decreased by a factor of 2.

So, the reaction is first order in Br₂.

3. Determine the overall rate order.

Since the reaction is second order in NO and first order in Br₂, the overall reaction rate is third order: $2+1=3$.

4. Determine the rate law.

Reaction rate = $k[A]^m[B]^n$ where $m=2$ and $n=1$

Reaction rate = $k[\text{NO}]^2 [\text{Br}_2]$.

Problem 4

EXAMPLE 9.3: DETERMINING THE RATE LAW FROM EXPERIMENTAL MEASUREMENTS

Determine the rate law and the rate constant for the reaction of carbon monoxide gas with nitrogen dioxide to give carbon dioxide and nitrogen monoxide from the following measurements:

Experiment	[CO] (M)	[NO ₂](M)	Rate (M/h)
(1)	5.10×10^{-4}	0.350×10^{-4}	3.4×10^{-8}
(2)	5.10×10^{-4}	0.700×10^{-4}	6.8×10^{-8}
(3)	1.02×10^{-3}	0.350×10^{-4}	6.8×10^{-8}

Solution 3

1. Determine the reaction order for NO_2 .

In experiments 1 and 2: the CO concentration is constant and the concentration of NO_2 is doubled. The reaction rate also doubled.

This means that the reaction is first order in NO_2 .

2. Determine the reaction order for CO.

In experiments 2 and 3: the concentration of NO_2 is constant and the concentration of CO is doubled. The reaction rate is also doubled.

The reaction is first order in CO.

3. Determine the reaction rate law.

Since the reaction is first order in CO and first order in NO_2 , the reaction rate law is:

$$\text{Reaction rate} = k[\text{CO}][\text{NO}_2]$$

4. Determine the value of the rate constant.

$$\text{Rate} = k[\text{CO}][\text{NO}_2]$$

$$k = \frac{\text{Rate}}{[\text{CO}][\text{NO}_2]} = \frac{(3.4 \times 10^{-8} \text{ M} \cdot \text{h}^{-1})}{(5.10 \times 10^{-4} \text{ M})(0.350 \times 10^{-4} \text{ M})} = 1.9 \text{ M}^{-1} \cdot \text{h}^{-1}$$

Solution 4: Alternate Method

$$\frac{\text{Rate (Experiment 2)}}{\text{Rate (Experiment 1)}} = \frac{6.8 \times 10^{-8} \text{ M h}^{-1}}{3.4 \times 10^{-8} \text{ M h}^{-1}} = \frac{k[\text{CO}]^m[\text{NO}_2]^n}{k[\text{CO}]^m[\text{NO}_2]^n}$$

$$2 = \frac{k[\text{CO}]^m[\text{NO}_2]^n}{k[\text{CO}]^m[\text{NO}_2]^n}$$

$$2 = \frac{\cancel{k}[5.1 \times 10^{-4}]^m[0.7 \times 10^{-4}]^n}{\cancel{k}[5.1 \times 10^{-4}]^m[0.35 \times 10^{-4}]^n}$$

$$2 = [2]^n$$

$$\log 2 = n \log 2$$

$$n = 1$$

**The reaction is first order
with respect to NO₂**

Solution 4: Alternate Method-Mathematical Procedure

$$\frac{\text{Rate (Experiment 2)}}{\text{Rate (Experiment 1)}} = \frac{6.8 \times 10^{-8} \text{ M h}^{-1}}{3.4 \times 10^{-8} \text{ M h}^{-1}} = \frac{k[\text{CO}]^m[\text{NO}_2]^n}{k[\text{CO}]^m[\text{NO}_2]^n}$$

$$2 = \frac{k[\text{CO}]^m[\text{NO}_2]^n}{k[\text{CO}]^m[\text{NO}_2]^n}$$

$$2 = \frac{\cancel{k}[5.1 \times 10^{-4}]^m[0.7 \times 10^{-4}]^n}{\cancel{k}[5.1 \times 10^{-4}]^m[0.35 \times 10^{-4}]^n}$$

$$2 = [2]^n$$

$$\log 2 = n \log 2$$

$$n = 1$$

Solution 3: Alternate Method (continued)

$$\frac{\text{Rate (Experiment 3)}}{\text{Rate (Experiment 1)}} = \frac{6.8 \times 10^{-8} \text{ M h}^{-1}}{3.4 \times 10^{-8} \text{ M h}^{-1}} = \frac{k[\text{CO}]^m[\text{NO}_2]^n}{k[\text{CO}]^m[\text{NO}_2]^n}$$

$$2 = \frac{k[\text{CO}]^m[\text{NO}_2]^n}{k[\text{CO}]^m[\text{NO}_2]^n}$$

$$2 = \frac{\cancel{k}[1.02 \times 10^{-3}]^m[\cancel{0.35} \times 10^{-4}]^n}{\cancel{k}[5.1 \times 10^{-4}]^m[\cancel{0.35} \times 10^{-4}]^n}$$

$$2 = [2]^m$$

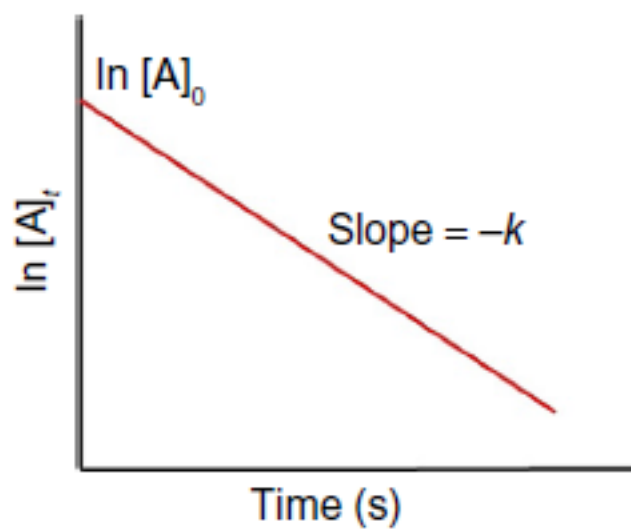
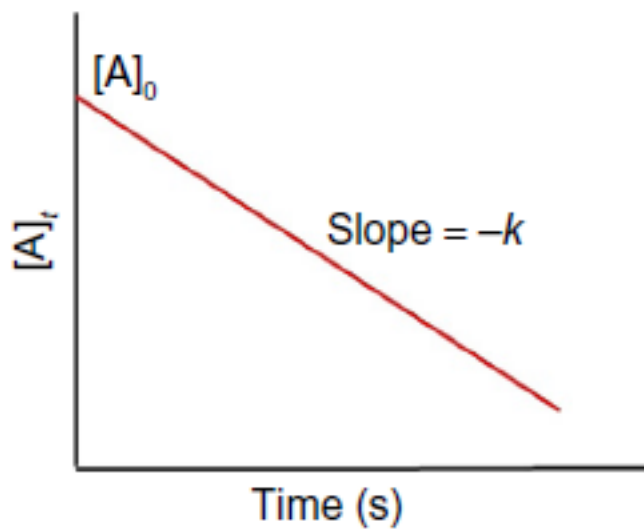
$$\log 2 = m \log 2$$

$$m = 1$$

**The reaction is first order
with respect to CO**

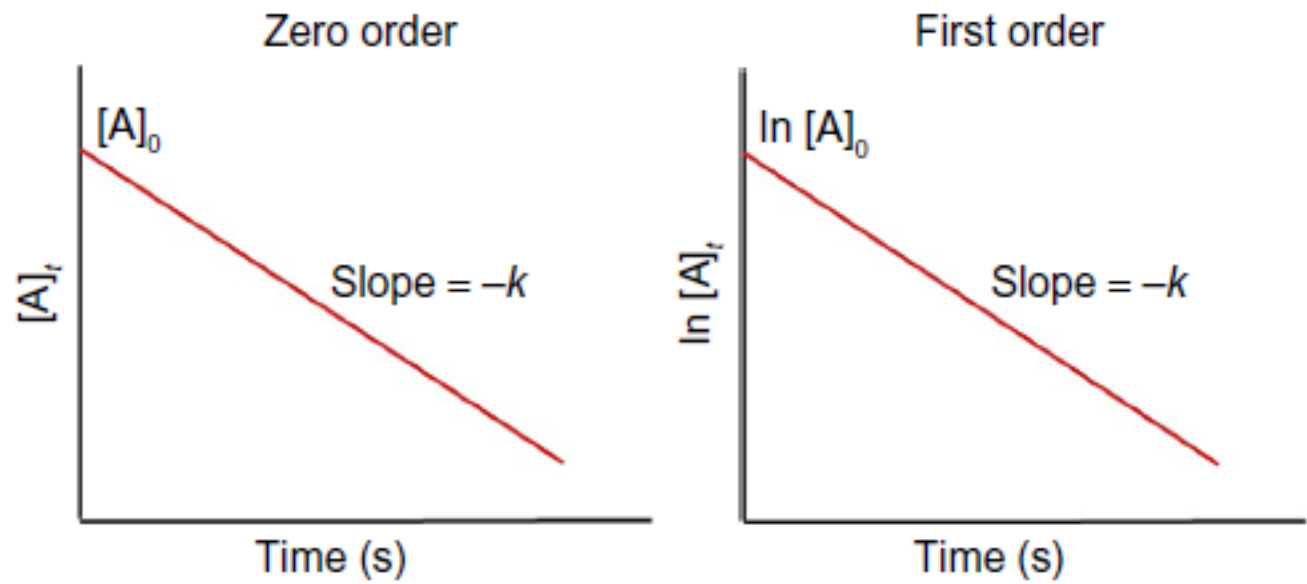
Problem 5

Which graph belongs to first order reaction?



Solution 5

FIG. 9.1 The plot of $[A]_t$ as a function of time for a zero order reaction (left) and $\ln[A]_t$ as a function of time for a first order reaction (right).



$$\int d[A] = -k \int dt$$
$$[A]_t = [A]_0 - kt$$

$$\int \frac{d[A]}{[A]} = -k \int dt$$
$$[A]_t = [A]_0 e^{-kt}$$
$$\ln [A]_t = \ln [A]_0 - kt$$

Problem 6

EXAMPLE 9.6: DETERMINING THE REACTION TIME FROM THE REACTION HALF LIFE

How much time will it take for the concentration of a reactant in a first order reaction to fall from 8 to 0.5M if the half life is of 10s?

Solution 6

1. Determine the concentration reduction factor.

$$0.5 \text{ M}/8 \text{ M} = 1/16$$

2. Determine the number of half lives.

After n half lives, the initial reactant concentration is reduced by $1/2^n$

$$1/2^n = 1/16$$

$$n = 4 \text{ half lives}$$

3. Determine the time elapsed.

$$4 \text{ half lives} = 4 \times 10\text{s} = 40\text{s}$$